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EXAMINING THE CRITERIA FOR PATENTING AI-BASED INVENTIONS AND THE EVOLVING STANDARDS IN DIFFERENT JURISDICTIONS

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ABSTRACT

With the rapid development and expansive integration of Artificial Intelligence (AI) across diverse sectors such as healthcare, law, finance, education, and creative industries, a fundamental shift has emerged in how societies interact with technology. This technological surge has compelled a critical re-examination of various legal doctrines, most prominently within the field of intellectual property (IP) law. In particular, patent law has encountered profound theoretical and practical challenges in light of machine-generated innovations. This dissertation sets out to explore, in a detailed and comparative manner, the evolving standards for patentability of AI-based inventions and how traditional patent criteria are being redefined—or resisted—across key jurisdictions.

A central issue explored in this study is the question of inventorship. Patent law, as it currently stands in most legal systems, relies heavily on the notion of human creativity and intention, framing inventorship around individuals with mental conception of an idea. However, the rise of AI systems capable of autonomous problem-solving and novel output generation presents an unprecedented challenge to this anthropocentric model. When an AI system, devoid of legal personality, generates a potentially patentable invention without human intervention, existing statutory provisions fall short of offering a viable legal framework. This study critically examines the conceptual and legislative boundaries of human inventorship in the context of algorithmically generated inventions.[i]

Equally significant is the examination of the substantive conditions of patentability—namely, novelty, inventive step (non-obviousness), and industrial applicability—as applied to AIgenerated outputs. Given that AI systems often rely on pre-existing data sets and machine learning models to derive outcomes, the delineation between computational output and creative inventiveness becomes increasingly blurred. This research investigates whether current legal doctrines are equipped to evaluate the originality and technical merit of AI inventions using existing standards, or whether these criteria necessitate conceptual overhaul.

The dissertation undertakes a comparative analysis of patent law responses from major jurisdictions, including the United States, the United Kingdom, the European Union, and India. Each jurisdiction presents distinct approaches to defining inventorship, assigning legal rights, and interpreting patentability in the AI context. These legal variations reveal a broader international ambivalence regarding AI's legal status and the scope of its creative agency.

The findings of this research reveal a significant lacuna in global patent law when it comes to acknowledging the evolving nature of technological innovation. AI is increasingly functioning not just as a tool, but as a semi-autonomous contributor to inventive processes. Absent legislative reform, this legal void may disincentivize innovation and obstruct the full potential of AI integration into society's creative and scientific fabric. This dissertation concludes by proposing actionable policy reforms and legal adaptations to better accommodate AI-driven inventorship while maintaining the fundamental principles of the patent system. Recommendations include reinterpreting existing definitions of inventorship to include AI-assisted contributions, introducing sui generis rights for machine-generated inventions, and fostering international harmonization of legal standards to ensure both innovation and legal certainty in the AI era.

CHAPTER 1

INTRODUCTION

"I alone cannot change the world, but I can cast a stone across the waters to create many ripples."
— Mother Teresa

1.1 THE PERSPECTIVE

Artificial Intelligence (hereinafter referred to as AI) has emerged as one of the most transformative and disruptive innovations of the mid-twenty-first century. This is not merely due to its ability to mimic human learning and operate without direct human oversight, but because of its inherent potential to process experiences and act autonomously, akin to living intelligence. While the concept of AI has existed for more than six decades, its widespread and far-reaching impact is a more contemporary development.[1]

[1] L Bently and B Sherman, (2014), "Intellectual Property Law", 4th ed., Oxford University Press, London, UK, p.43, ISBN-9780199645558.

The influence of AI is increasingly visible in day-to-day life, ranging from game-playing bots and autonomous vehicles to algorithmic recommendations on digital platforms, AI-assisted medical diagnoses, facial recognition systems, and even the generation of artificial images. These are only a few illustrations of how AI is being embedded into our daily environments. Various sectors—including the legal industry, air traffic management, healthcare, the Internet of Things (hereinafter referred to as IoT), and especially intellectual property (hereinafter referred to as IP)—have witnessed the extensive integration of AI-driven technologies. In fact, AI has become integral to the operation and maintenance of both tangible and digital infrastructure, including components governed by IP rights. Due to AI's autonomous functionality and learning capability independent of human input, significant challenges have emerged across various IP branches, especially in the domain of copyright.[2]

The intersection of AI and IP has become one of the most debated legal and technological issues of the 21st century. The core of this debate lies in AI's ability to be deployed across a multitude of sectors to perform tasks traditionally carried out by human beings, thereby aiming to enhance efficiency or reduce human effort.

AI introduces the concept of a system capable of independent thought, a faculty traditionally reserved for human cognition. It allows machines and programs to interpret their environment, evaluate options, and produce rapid, precise responses. As a result, AI systems are now capable of executing tasks that were once considered exclusive to human intelligence.[3]

A recent report from the United States (hereinafter referred to as the US) stated that the global AI market, which stood at 2.42 billion USD in 2017, is projected to grow to approximately 59.7 billion USD by 2025. Furthermore, a forecast by

PricewaterhouseCoopers (hereinafter referred to as PwC),[4] a UK-based multinational professional services firm, anticipates that AI could potentially add a staggering 15.7 trillion USD to the global economy by the year 2035. At this point in technological evolution, nearly every facet of human life is being touched by AI, including the legal realm. The European

Patent Office has gone so far as to categorize AI as the cornerstone of the "Fourth Industrial Revolution." Looking ahead, it is likely that humanity will need to address new societal challenges brought about by AI, challenges that will inevitably impact the legal frameworks that govern us.

AI technologies are already changing the foundation of how industries operate and innovate. For example, businesses are now deploying AI to optimize workflows, reduce overhead costs, and enhance decision-making through predictive analysis. Governments are incorporating AI to improve public administration, automate routine tasks, and monitor data at a scale previously unthinkable. As AI systems evolve, their capabilities in

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[2] Arthur R. Miller, 1993, 'Copyright protection for computer programs, databases, and computer- generated works; is anything new since CONTU?', Harvard Law Review, March, vol. 106, no. 5, pp. 977-1073

[3] Jeremy A. et.al., 2018, Research handbook on the law of AI, Edward Elgar Publishing Limited, Massachusetts, USA, p.519, ISBN- 9781786439048.

[4] PricewaterhouseCoopers, Sizing the Prize (2017) 22.

creating new and innovative solutions—whether software-generated or physical inventions—are becoming increasingly sophisticated.

The growing capability of AI to autonomously generate ideas and solutions has compelled legal scholars, governments, and international bodies to reconsider the foundations of current intellectual property laws. While AI was initially limited to supporting roles, such as facilitating analysis or enhancing efficiency in the invention process, it now holds the

potential to act as an independent contributor to innovation. This shift raises a fundamental legal and philosophical question: can a machine or algorithm be regarded as an "inventor" under existing legal frameworks?

This very question has garnered the attention of patent offices and courts across multiple jurisdictions. The introduction of AI-generated inventions has sparked debates on the allocation of patent rights, inventorship, and ownership, particularly when the traditional human-centric model of innovation is no longer applicable. There is increasing concern that the current patent laws may not be equipped to accommodate inventions produced entirely or partly by AI systems.

At the core of patent law lies the principle that inventions must be novel, involve an inventive step, and be capable of industrial application. These principles were traditionally drafted with human inventors in mind, assuming human ingenuity as the source of creativity. However, AI systems, through deep learning and neural network algorithms, are now capable of analyzing vast datasets, identifying patterns, and formulating novel solutions with minimal or no human intervention. This evolution challenges the conventional criteria of what constitutes an "inventor" and whether such AI-generated results meet the threshold for patentability.[5]

Moreover, the absence of clear global standards on this issue has led to conflicting interpretations across jurisdictions. For instance, the European Patent Office (hereinafter referred to as EPO), the United States Patent and Trademark Office (hereinafter referred to as USPTO), and the United Kingdom Intellectual Property Office (hereinafter referred to as UKIPO) have all dealt with applications listing an AI system, DABUS (Device for the Autonomous Bootstrapping of Unified Sentience), as the inventor. In each of these instances, the applications were rejected on the grounds that inventorship is limited to natural persons.

This legal uncertainty not only impacts the inventors and organizations involved in AI development but also affects the broader innovation ecosystem. If innovators are unsure whether their AI-generated creations are eligible for patent protection, they may be less likely to disclose them through patent filings. Instead, they

[5] Eliza Mik, 2021, AI as a Legal Person, In: Jyh-An Lee, et.al., Artificial Intelligence and Intellectual Property, Oxford University Press, Oxford, United Kingdom, 1st ed., p. 430 ISBN 978-0- 19- 887094-4.

might opt for trade secrets, which could reduce the public availability of knowledge and stifle collaborative progress in the tech community.

Hence, there is a pressing need to reevaluate the parameters of patent law to ensure it remains inclusive and adaptable to technological advancements, particularly AI. It becomes imperative to assess whether the current frameworks can accommodate inventions generated by machines or whether new legislative interventions are required to recognize the evolving nature of inventorship.

1.2 ARTIFICIAL INTELLIGENCE: AN OVERVIEW

Artificial Intelligence (AI) is a multidisciplinary field that focuses on the creation of intelligent systems capable of performing tasks traditionally requiring human intelligence. These tasks include reasoning, problem-solving, decision-making, understanding natural language, visual perception, and learning from experience. Initially explored as a subset of computer science, AI has evolved through significant technological strides, branching into domains like robotics, machine learning, deep learning, natural language processing, and expert systems.

The roots of AI can be traced back to the mid-20th century, when pioneers like Alan Turing laid the conceptual groundwork. Turing's famous question—"Can machines think?"—posed in his 1950 paper, "Computing Machinery and Intelligence," introduced the foundational concept of machine intelligence.[6] Over the decades, AI research has advanced significantly, especially with the advent of increased computational power, the availability of big data, and the development of advanced algorithms.

In contemporary usage, AI systems can be broadly categorized into two types: **narrow AI** and **general AI**. Narrow AI (also known as weak AI) refers to systems designed to perform specific tasks, such as facial recognition, language translation, or autonomous navigation. These systems operate under limited parameters and cannot perform tasks outside their predefined functions. In contrast, general AI (or strong AI) aims to replicate the full range of human cognitive abilities, allowing machines to learn and adapt to various tasks autonomously—though this form remains largely theoretical at present.

The primary driving force behind modern AI advancements is **machine learning**, a technique that enables machines to improve their performance based on experience. Machine learning systems are trained on large datasets, using statistical models to identify patterns and make predictions. A more sophisticated subset of this is **deep learning**, which employs neural

networks with multiple layers to simulate the human brain's functioning. These models are particularly effective in handling unstructured data like images, audio, and natural language.

AI technologies have increasingly found application in fields as diverse as healthcare, finance, education, manufacturing, transportation, and law. For instance, AI-driven diagnostic tools can identify diseases from

[6] Alan Turing, 'Computing Machinery and Intelligence' (1950) 59 Mind 433, 436.

medical imaging with high precision, while in finance, AI is used to detect fraudulent transactions and predict market trends. In the legal domain, AI tools assist in contract review, legal research, and predictive analytics, thus streamlining many traditionally manual processes.

As AI becomes more embedded in core decision-making processes, it raises significant ethical, social, and legal concerns. These include issues around data privacy, algorithmic bias, accountability, transparency, and the displacement of human labor. From a legal standpoint, the capacity of AI to generate content, make autonomous decisions, or invent independently prompts a reevaluation of intellectual property rights and the definition of authorship and inventorship.

The capabilities of AI systems to not only assist but potentially replace human ingenuity in certain tasks challenge the traditional frameworks within which the law operates. Particularly in the context of patent law, where the idea of an "inventor" has historically been tied to human consciousness and creativity, AI's involvement disrupts these long-held assumptions.

Consequently, understanding the fundamental concepts and operational dynamics of AI becomes critical in assessing its compatibility with existing legal structures. It sets the stage for examining how patent regimes across various jurisdictions are responding to the emergence of AI-generated inventions and whether new legal paradigms are necessary to govern this evolving landscape.

1.3 UNDERSTANDING PATENTS AND THE LEGAL NOTION OF INVENTORSHIP

Patents are a subset of intellectual property rights that provide inventors with exclusive rights over their inventions for a specified duration, typically 20 years from the date of filing. A patent grants its holder the legal authority to prevent others from making, using, selling, or distributing the patented invention without consent. This legal exclusivity serves as a powerful incentive for innovation by ensuring that inventors can capitalize on their creations, thereby encouraging continued investment in research and development.[8]

The modern patent system is grounded in a delicate balance: it rewards inventors for their ingenuity while ensuring that the public eventually gains access to technological advancements. In exchange for exclusive rights, inventors must disclose their inventions in sufficient detail so that others skilled in the art can replicate them. This mechanism ensures that technological progress becomes part of the public domain once the patent expires.

The legal framework governing patents varies across jurisdictions but generally adheres to a set of common criteria that an invention must fulfill to be eligible for patent protection. These include:

1. **Novelty**: The invention must be new, meaning it should not form part of the prior art—i.e., it should not have been publicly disclosed before the filing date.
2. **Inventive Step or Non-Obviousness**: The invention should involve an inventive step that would not be obvious to a person having ordinary skill in the art.

3. **Industrial Applicability or Utility**: The invention must be capable of industrial application or provide a specific, substantial, and credible utility.

4. **Patentable Subject Matter**: The invention must fall within the domain of patentable subject matter, which varies by jurisdiction.

Among these, the concept of **inventorship** is central to patent law, as it establishes the legal origin of the right to apply for and own a patent. Traditionally, the **"inventor"** is understood as a natural person who contributes to the conception of the claimed invention. The legal notion of inventorship is intimately tied to the human mind, creativity, and intention. This human-centric understanding has been reinforced by numerous judicial decisions and legal doctrines across jurisdictions.

For example, under U.S. patent law, only natural persons can be recognized as inventors. The Federal Circuit, in *University of Utah v. Max-Planck-Gesellschaft zur Förderung der Wissenschaften e.V.*, held that "conception is the touchstone of inventorship," which must involve a mental act performed by a human being.[7] Similarly, the European Patent Convention (EPC) implicitly assumes a human inventor, requiring the designation of the inventor in the application process, and the Indian Patents Act mandates that the inventor be a person, reinforcing the anthropocentric bias.

Determining inventorship is not merely an academic concern—it has significant legal implications. Improper designation of inventors can lead to the invalidation of patents, disputes over ownership, and infringement claims. Moreover, inventorship affects the rights of employers, employees, research institutions, and collaborative ventures, often dictating how profits are distributed and control is exercised over the invention.

With the advent of Artificial Intelligence, especially systems capable of generating novel solutions, the conventional understanding of inventorship is being challenged. AI systems, particularly those that learn and adapt independently, blur the lines between tools and creators. Can an AI be an inventor if it autonomously develops a new and non-obvious solution to a technical problem? Should the creator or owner of the AI system be credited instead? These questions highlight the growing tension between evolving technological capabilities and static legal doctrines.

In summary, the traditional notion of inventorship, firmly rooted in the idea of human creativity and legal personhood, is now under scrutiny. As AI increasingly contributes to, or even independently generates, innovative solutions, there is a pressing need to reconsider whether existing legal frameworks remain adequate or whether new categories of inventorship must be developed to accommodate non-human creators.

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[7] University of Utah v Max-Planck-Gesellschaft 734 F3d 1315, 1323 (Fed Cir 2013).

1.4 INTERSECTION OF AI AND PATENT LAW

The rapid integration of Artificial Intelligence into a diverse range of industries has fundamentally altered the landscape of innovation. AI systems today are not merely computational tools used to assist human inventors but are increasingly capable of generating ideas, solutions, and even entire inventions independently. This development compels a reevaluation of the foundational principles of patent law, especially regarding the criteria for inventorship and patent eligibility.

The convergence of AI and patent law introduces several unprecedented legal challenges. Traditionally, patents have been granted for inventions that are the product of human ingenuity. The legal definitions of terms such as "inventor" and "conception" have consistently presupposed human involvement.[8] However, the growing capacity of AI systems

to produce novel, useful, and non-obvious outcomes without human guidance disrupts this assumption. In such cases, it becomes increasingly difficult to determine whether the output of an AI system qualifies as a patentable invention and, if so, who should be recognized as the rightful inventor.

One of the most prominent examples demonstrating this tension is the **DABUS (Device for the Autonomous Bootstrapping of Unified Sentience)** case. In this case, the AI system DABUS, developed by Dr. Stephen Thaler, was named as the inventor on patent applications filed in multiple jurisdictions, including the United States, United Kingdom, European Union, and South Africa. The applications claimed that DABUS had autonomously generated the inventions without human input. While South Africa granted the patent recognizing DABUS as an inventor—the first country to do so—most other jurisdictions rejected the application on the grounds that only a natural person can be legally recognized as an inventor.[9]

This case triggered an international debate about whether existing patent frameworks are equipped to handle AI-generated inventions. Courts and patent offices have emphasized that current laws do not allow for non-human entities to be named as inventors, reflecting the longstanding assumption that only human beings possess the requisite mental faculties for "conception." Despite technological advancements, legal systems across jurisdictions have maintained this position, citing the need for legal accountability, moral rights, and the practicalities of legal ownership and enforcement—all of which presuppose human inventors.

Yet, this conservative legal stance risks discouraging innovation. AI-generated inventions, particularly those created without human input, may not qualify for patent protection under existing frameworks. This could lead to the under-disclosure of AI-generated inventions, reliance on trade secrets, or legal ambiguities in ownership—all of which could stifle technological progress and investment in AI research.

Furthermore, AI challenges the **inventive step** requirement of patent law. Patent offices must evaluate whether the output of an AI system is truly non-obvious. However, AI systems often operate in ways that are opaque

[8] EPO, Guidelines for Examination (2023) G-II 3.1.

[9] DABUS Patent (ZA 2021/03242) (South Africa Patent Office).

even to their creators, making it difficult to assess how an invention was derived or whether it meets the threshold of non-obviousness. This raises concerns about how patent examiners can fairly and consistently evaluate AI-generated

inventions, particularly when AI systems are trained on vast datasets and operate in unpredictable ways.

In light of these developments, legal scholars and policymakers are increasingly calling for reforms to the existing patent system. Proposals include the creation of new legal categories for AI-assisted or AI-generated inventions, recognition of the AI system's owner or developer as a proxy inventor, or the establishment of sui generis protection regimes tailored to AI. Each of these proposals, however, carries complex implications regarding ownership rights, attribution, liability, and international harmonization.

In conclusion, the intersection of AI and patent law presents both a challenge and an opportunity. While current legal frameworks remain anchored in human-centric notions of inventorship, the increasing autonomy of AI in the innovation process demands a forwardlooking re-examination of these doctrines. Whether through judicial reinterpretation, legislative amendment, or international consensus-building, the legal system must evolve to remain aligned with the realities of AI-driven innovation.

1.5 RESEARCH OBJECTIVES

This dissertation aims to undertake a comprehensive examination of the legal framework surrounding the patentability of inventions involving Artificial Intelligence, with particular attention to the criteria applied across various jurisdictions. As AI continues to advance rapidly, it has prompted critical debates about whether existing patent systems are equipped to recognize and protect such inventions adequately. This research, therefore, seeks to identify, evaluate, and clarify the evolving standards of patent law as they intersect with AI technologies.

The central objective is to assess whether the traditional patent eligibility criteria—such as novelty, inventive step (or non-obviousness), and industrial applicability—are sufficient and appropriate for inventions generated by or with the assistance of AI. In doing so, the study will consider not only the doctrinal aspects of patent law but also the technological and philosophical questions that AI presents regarding creativity, authorship, and legal accountability.

A critical component of this analysis involves comparing the approaches of major patent jurisdictions, including the United States, European Union, United Kingdom, India, China, Japan, and South Korea. Each of these regions has responded differently to the challenges posed by AI-generated inventions, and these differences provide valuable insight into the broader debate over how to modernize global intellectual property frameworks.

Furthermore, this dissertation seeks to:

- Investigate the legal implications of attributing inventorship to AI systems and whether alternative mechanisms—such as recognizing the developer, owner, or user of the AI as the legal inventor—are viable or desirable.
- Evaluate recent case law and decisions by patent offices around the world that deal with AI-related patent applications, such as the DABUS case, to determine the prevailing trends and judicial reasoning applied.
- Examine the philosophical underpinnings of inventorship and whether the concept can or should be expanded to include non-human entities.
- Propose recommendations for future legal reform that balance the need to protect innovation with the unique characteristics of AI technologies, while ensuring legal certainty and international harmonization.

Ultimately, the research aims to contribute to the academic and legal discourse by offering a structured, comparative, and forward-thinking analysis of how AI is reshaping patent law. It aspires to assist legislators, policymakers, legal practitioners, and inventors in navigating the complexities of this evolving field and in developing responsive legal frameworks that foster innovation without compromising foundational legal principles.

1.6 RESEARCH QUESTIONS

To provide a structured analysis of the intersection between Artificial Intelligence and patent law, this dissertation is guided by the following core research questions:

1. **Are existing legal standards for patentability, such as novelty, inventive step, and industrial applicability, sufficient to accommodate the unique nature of AIgenerated inventions?**
   This question seeks to assess whether traditional patent criteria are capable of recognizing innovations created with or by AI technologies, or whether such standards need adaptation to reflect the complexities of modern, machine-led innovation.

2. **How do leading jurisdictions differ in their interpretation and application of patentability standards to AI-related inventions?**
   Through a comparative study, this question examines the legislative and judicial treatment of AI inventions in regions such as the United States, European Union, India, China, and Japan, among others, to determine the extent of divergence or harmonization in global patent regimes.

3. **Who should be recognized as the inventor when an invention is autonomously generated by an AI system, and what legal frameworks support or oppose such recognition?**
   This question explores the controversial issue of AI inventorship—whether machines can or should be named as inventors under current patent laws and, if not, who among the human stakeholders (developer, owner, user) should hold that status.[35]

4. **What are the philosophical and legal implications of granting inventorship or patent rights to non-human entities?**

   Beyond the practical, this question dives into the ethical and jurisprudential dimensions of attributing legal personhood or rights to AI systems, including whether such a move aligns with established principles of intellectual property law.[36]

5. **What reforms or policy changes are necessary to ensure that patent law remains effective, fair, and conducive to innovation in the context of AI?**

   This final question addresses the future of patent systems, identifying gaps in existing laws and proposing potential reforms that maintain legal coherence while accommodating the growing role of AI in the innovation ecosystem.[37]

Each of these questions plays a vital role in shaping the analysis throughout this dissertation. Together, they form the backbone of the inquiry into how patent law must evolve—or resist evolution—in response to the technological disruptions introduced by artificial intelligence.

1.7 RESEARCH METHODOLOGY

The research methodology employed in this dissertation is designed to provide a comprehensive and systematic exploration of the legal issues surrounding AI-generated inventions, with a specific focus on patent law. The approach combines doctrinal research with comparative and analytical methods, supported by both primary and secondary sources.

The methodology is structured as follows:

1. **Doctrinal Research:**

   The primary method of research is doctrinal analysis, which involves examining existing legal texts, statutes, case law, and authoritative legal commentary. This methodology will be used to explore the current legal framework surrounding patent law, with a particular emphasis on the treatment of AI-generated inventions across different jurisdictions. Legal texts such as national patent laws, international treaties, and relevant case law will be critically analyzed to assess their capacity to address the challenges posed by AI in the patent system.

2. **Comparative Analysis:**

   Given the global nature of the issue at hand, the research also adopts a comparative legal methodology. This involves analyzing how various legal systems, such as those of the United States, European Union, India, China, and Japan, approach the patentability of AI-generated inventions. By comparing these jurisdictions, the dissertation will highlight the similarities and differences in their treatment of AI within patent law, ultimately identifying best practices and areas where reform is needed.

3. **Case Study Analysis:**

   To further understand how patent law has responded to AI-generated inventions, the dissertation will examine case studies involving AI technology and patenting practices. This includes key case law, such as the landmark decisions in the United States and Europe regarding the patenting of inventions created by AI systems. These case studies will help illustrate how courts and patent offices have interpreted existing laws in the context of AI innovations and the legal implications of these rulings.

4. **Interviews with Experts:**
   To complement the legal research, the dissertation will also incorporate insights from industry experts, legal practitioners, and policymakers. Semi-structured interviews will be conducted with professionals in the fields of intellectual property law, AI technology, and patent law. These interviews will provide practical perspectives on the challenges and potential solutions to the issues surrounding AI-generated patents.

5. **Critical Legal Theory:**
   The dissertation will incorporate critical legal theory to explore the philosophical and ethical dimensions of patenting AI-generated inventions. This theoretical lens will examine the broader implications of attributing inventorship or ownership of inventions to non-human entities, considering whether such a shift aligns with or challenges established legal norms and principles.

6. **Secondary Source Research:**
   In addition to primary sources, the research will utilize secondary sources, including academic articles, books, reports, and commentaries, to gain a broader understanding of the subject matter. These sources will provide context, historical background, and current trends in patent law as it pertains to AI, offering a well-rounded perspective on the issues being studied.

7. **Doctrinal Synthesis:**
   Finally, a doctrinal synthesis of the information gathered through the aforementioned methods will be carried out. This synthesis will aim to draw conclusions regarding the current adequacy of patent law in addressing AI-generated inventions, and propose recommendations for legal reform where necessary. The ultimate goal is to present a clear and well-supported argument on how patent law should evolve in response to the rise of artificial intelligence.

By combining these diverse research methodologies, the dissertation aims to offer a nuanced and detailed analysis of the intersection between AI and patent law, providing valuable insights for scholars, practitioners, and policymakers.

1.8 STRUCTURE OF THE DISSERTATION

This dissertation is organized into distinct chapters, each focusing on a specific aspect of the intersection between artificial intelligence (AI) and patent law. The structure of the dissertation is as follows:

Chapter 1: Introduction

The introductory chapter outlines the fundamental issues surrounding the patentability of AIgenerated inventions. It provides an overview of artificial intelligence, its capabilities, and its relevance to the legal field, particularly in relation to intellectual property (IP). The chapter also introduces the research objectives, methodology, and the importance of the study in addressing the emerging legal challenges in patent law. It sets the stage for the subsequent indepth analysis by discussing the scope, significance, and research questions that the dissertation seeks to answer.

Chapter 2: The Evolution of Artificial Intelligence and its Impact on Patent Law This chapter explores the evolution of AI technology and its growing influence on various sectors, with a particular focus on its application in the field of intellectual property. It traces the history of AI development, highlighting significant milestones, and examines how AI has started to influence patent law. The chapter also investigates the key differences between traditional inventions and those created autonomously by AI systems. It will discuss the various challenges that AI poses to the current patent regime, particularly in terms of inventorship, ownership, and legal accountability.

Chapter 3: Legal Framework for Patenting Inventions

The third chapter delves into the existing legal framework for patenting inventions, providing a detailed examination of patent laws across key jurisdictions, including the United States, European Union, and India. It covers the fundamental principles of patent law, including novelty, inventive step, and industrial applicability. The chapter will critically analyze the applicability of these traditional patent criteria to AI-generated inventions and discuss the legal complexities surrounding the issue of authorship, with a focus on AI as a potential inventor.

Chapter 4: Comparative Analysis of Patent Laws on AI-generated Inventions This chapter conducts a comparative analysis of how different countries and legal systems address the issue of AI-generated inventions. The focus will be on major jurisdictions such as the United States, the European Union, and China, examining how each has adapted or failed to adapt its patent laws to address the rise of AI. The chapter will highlight key case law, legislative approaches, and regulatory frameworks that reflect how various countries interpret the role of AI in the patent process. By contrasting these international approaches, the chapter will identify best practices and potential areas for reform in patent law.

Chapter 5: The Ethical and Philosophical Dimensions of AI-generated Inventions Chapter five explores the ethical and philosophical implications of patenting inventions created by AI. It examines the moral questions surrounding AI as an inventor and the potential consequences for human creativity and innovation. The chapter considers whether AI can truly be considered an "inventor" in the traditional sense and explores the broader societal implications of granting patent rights to non-human entities. It also considers the ethical implications of awarding intellectual property rights to AI-generated works, addressing concerns regarding the nature of ownership, credit, and accountability.

Chapter 6: Case Studies on AI and Intellectual Property

This chapter presents detailed case studies of AI-generated inventions that have been subjected to patent application processes in various jurisdictions. The case studies will focus on specific examples, such as AI-generated artworks, music, and literary works, and analyze how these cases have been handled by patent offices and courts. It will discuss the outcomes of these cases and the legal arguments that have been made regarding the patentability of AIgenerated inventions. Through these case studies, the chapter will illustrate the real-world challenges and legal uncertainties surrounding AI and IP.

Chapter 7: Challenges and Legal Implications of AI-generated Inventions This chapter identifies and discusses the key legal challenges posed by AI-generated inventions, particularly in the context of intellectual property law. It examines issues such as the definition of inventorship, ownership of AI-generated works, and the capacity of existing patent laws to adequately address these challenges. The chapter also explores potential conflicts between traditional patent laws and the reality of AI-generated inventions, offering a critical analysis of whether current legal frameworks can evolve to accommodate these technological advancements.

Chapter 8: Proposals for Reform and Legal Recommendations

The final chapter of the dissertation offers proposals for legal reform to address the challenges presented by AI-generated inventions. Based on the analysis presented in the previous chapters, this section provides recommendations for changes in patent law that would better accommodate AI and its role in the creative and inventive process. It discusses potential legislative amendments, regulatory reforms, and international cooperation efforts that could ensure the effective protection of AI-generated inventions while maintaining a balance between innovation, competition, and intellectual property rights.

Chapter 9: Conclusion

The conclusion summarizes the findings of the research, reiterating the main arguments and insights developed throughout the dissertation. It offers a final evaluation of the adequacy of current patent laws in dealing with AI-generated inventions and reflects on the need for reform in light of technological advancements. The conclusion also highlights the potential future trajectory of AI in the legal domain and the steps necessary to ensure that the law evolves in tandem with technological progress.

CHAPTER 2

CONCEPTUAL ANALYSIS OF ARTIFICIAL INTELLIGENCE

1. INTRODUCTION

As indicated by the title of this dissertation, **Artificial Intelligence (hereinafter referred to as AI)** serves as the central theme around which the entire study is structured. This chapter offers an exploration of the evolution of AI, tracing its progression from ancient philosophical concepts to its current applications in today's digital era. It also delves into several significant aspects of AI, such as its historical trajectory, definitions, key components, and its expanding role in the contemporary world.

AI has emerged as one of the most groundbreaking technologies of the modern age, drawing the interest of scholars and researchers from diverse domains including computer science, law, ethics, and philosophy. No longer confined to the realm of science fiction, AI has become an integral part of everyday life, assisting humanity in solving critical global problems and enhancing the ease of daily living. However, as AI continues to evolve, it is likely that society will confront novel social challenges, prompting the need for adaptive and forward-thinking legal frameworks to effectively regulate its impact.

Much like the transformative influence of the steam engine and electricity in previous centuries, AI is reshaping modern society and is widely regarded as one of the most strategic technologies of the 21st century. The manner in which we engage with and regulate AI will have far-reaching consequences on the future of human life.

Since its conceptual inception, global perspectives on AI have been polarized. On one hand, some believe that AI has the potential to revolutionize societal structures for the better, elevating quality of life and economic productivity. On the other hand, there are concerns that AI may eventually reach a point where machines gain dominance over human agency. Acknowledging the benefits that AI can offer to society, this research undertakes a comprehensive examination of its various dimensions.

Historically, scientists observed that tasks requiring substantial computational effort placed significant strain on the human brain, which has inherent cognitive limitations. This realization inspired the idea of creating intelligent machines that could emulate human cognitive processes and perform such tasks efficiently. These visionary efforts eventually laid the foundation for the development of AI.

Broadly speaking, AI is a specialized field within computer science that focuses on developing systems with human-like cognitive abilities, including reasoning, learning, and decision-making. The rapid global advancement of AI technologies has resulted in their integration into a wide range of everyday applications. For instance, Apple's use of Siri to enhance the functionality and performance of its device[10]s, Google's predictive capabilities in Google Maps[11]. facial recognition previously utilized by

Facebook, and YouTube's recommendation algorithms—all exemplify how AI is now deeply embedded in daily technological experiences[12]. Remarkably, in 2017, Saudi

Arabia even granted citizenship to a humanoid robot named Sophia, symbolizing AI's growing cultural significance.

The integration of terms like "Deep Learning," "Machine Learning," and "Neural

Networks" into everyday language reflects how AI has become a defining element of the modern technological and cultural landscape. Today, nearly every sector—ranging from journalism and entertainment to art and architecture—is experiencing AI-driven transformation. Advanced AI systems are now capable of composing music, writing poems and articles, designing buildings, and even generating visual art—often surpassing the creative abilities of human beings in certain contexts.

[10] Bushnell, M. 2021, Artificial Intelligence Faceoff: Siri vs. Cortana vs. Google Assistant vs. Alexa, Business News Daily Stuff Online, 08 November

[11] ET Online, 2020, How Google Maps is using AI and Machine Learning to understand post-COVID traffic, The Economic Times: Tech, 04 September

[12] Bernard Marr, 2019, The Amazing Ways YouTube Uses Artificial Intelligence andMachine Learning, Forbs Online, 23 August

Moreover, innovations like AI-powered legal assistants, AI doctors, autonomous vehicles, AI-based judicial systems, intelligent surveillance mechanisms, AI-managed traffic control networks, and hyper-realistic AI-generated human faces, exemplify the expansive and transformative reach of this technology. These developments are only a glimpse of the profound ways AI is reshaping the world.[13]

Consequently, while AI is paving the way for the rise of new industries, it is simultaneously posing complex challenges to the existing intellectual property (IP) framework. Addressing these emerging concerns necessitates a detailed understanding of the various concepts surrounding AI, which this chapter aims to provide through systematic analysis and discussion.

2. PHASE WISE HISTORICAL DEVELOPMENT OF ARTIFICIAL INTELLIGENCE

Although Artificial Intelligence (AI) is widely regarded as a modern innovation, its conceptual roots can be traced back to ancient times. The early philosophers and scholars who speculated on the nature of human thought and logic could hardly have envisioned that their abstract ideas would one day evolve into what we now identify as AI. Historically, the development of AI has been characterized by a progressive attempt to simulate elements of human cognition and skill that can be modeled computationally.

The initial applications of AI were largely confined to formal disciplines such as theorem proving—domains that were far removed from the nuanced realities of human daily life. However, with advancements in expert systems, natural language processing, and robotics, AI has expanded its scope to encompass a broader range of human activities and behaviors.

The development of AI has not been without debate and differing perspectives. For some, AI represents the effort to recreate human intelligence artificially, while others interpret it as a mechanistic explanation of activities that resemble human behavior. These conflicting viewpoints have fueled both fascination and skepticism throughout AI's developmental history.

In light of these complexities, the researcher believes that revisiting the imaginative ideas of ancient thinkers can offer meaningful insights into the evolution of modern AI. To facilitate a clearer understanding, the historical trajectory of AI in this study is divided into two key phases. The first phase spans from ancient philosophical musings to the mid20th century, culminating in the formal coining of the term "Artificial Intelligence" by John McCarthy in 1956. The second phase encompasses the post-McCarthy era, marking the continued evolution of AI technologies into the modern age.

[13] Nicholas V. Findler, 2007, AI, Emeritus Voices, The Journal of the Emeritus College at ASU, 01 October

A chronological examination of these phases serves to clarify the development of AI and highlights how early conceptualizations gradually transformed into the sophisticated technological reality we witness today.

2.1 ANCIENT DEVELOPMENT OF ARTIFICIAL INTELLIGENCE

During antiquity, the Greek philosopher Aristotle (384–322 BC) introduced the concept of logic, which laid the groundwork for structured reasoning and ultimately influenced the development of Artificial Intelligence (AI)[14]. His pioneering ideas on logical frameworks became a foundational element for later advancements by scholars in both mathematics and philosophy, paving the way for AI's conceptual evolution.

In 1308, Catalan poet and philosopher Ramon Llull proposed a system of reasoning based on permutations in his work *Ars Magna*, aiming to represent logic through mechanical means. Several centuries later, in 1666, German polymath Gottfried Wilhelm Leibniz developed a machine that could process conceptual inputs—albeit within a limited scope—and authored *Dissertatio de Arte Combinatoria* ("On the Combinatorial Art"), where he posited that complex ideas are essentially combinations of simpler ones[15]. Leibniz's effort to formulate a universal symbolic language of logic ultimately laid the foundation for modern calculus and influenced the propositional logic systems central to contemporary AI research, despite the fact that he could not realize a complete language of reasoning.

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14 Geneviève Fieux-Castagnet and Gerald Santucci, 2019, AI: A Civilizational Challenge, A Generational Duty, Big Tech Géopolitique, September

15 16 Stuart J. Russell and Peter Norvig , 2021, Artificial Intelligence "A Modern Approach, Pearson Series in AI, Pearson Education, Inc., 4th Ed., p.41, ISBN-13: 978-0-13-461099-3.

Building upon these logical traditions, English mathematician George Boole introduced what is now known as Boolean algebra—a system of symbolic logic that remains crucial in computing. Boolean logic underpins modern digital circuit design, with statements like "A is true" or "A is true and B is false" forming the basis of logical gate operations in computer hardware.

In parallel, Boole's contemporary Charles Babbage designed the Analytical Engine, considered the earliest conceptual model of a mechanical computer. Although a fully operational version was never completed during his lifetime, Babbage's vision inspired future innovations. This culminated in the creation of the Atanasoff–Berry Computer (ABC) in the 1940s, the first digital programmable machine, which replaced mechanical computation with electronic processes.

By the late 1950s, several fully functional electronic computers had emerged, replacing earlier mechanical models and employing vacuum tube technology. These developments marked a significant milestone in computing history, establishing the digital computer as a platform upon which Artificial Intelligence could be built. The idea of constructing an "electronic brain"[14] gained momentum during this period, igniting the modern pursuit of machines capable of mimicking human intelligence.

2.2 MODERN DEVELOPMENT OF ARTIFICIAL INTELLIGENCE

The second phase in the evolution of Artificial Intelligence has experienced a series of fluctuations, including periods of stagnation where interest from both government bodies and private enterprises diminished significantly. This lack of enthusiasm led to a substantial decline in funding and research opportunities, a time commonly referred to as the "AI Winter" of the 1970s. Despite this downturn, dedicated researchers continued to explore and advance the field, undeterred by the lack of support. For the purpose of better understanding this phase, the progression of AI during this era can be broadly categorized into the following developmental stages:

- Incubation Period (Mid-17th century to 1930)
- Emergence of AI (1940s to 1950s)
- Flourishing Period (1950 to 1970)
- First AI Winter (1970s to 1980s)
- AI Renaissance (1980 to 1987)
- Second AI Winter (1987 to 1993)
- AI Boom (1993 to 2016)
- AI Breakthrough Era (2016 to present)

2.2.1 THE INCUBATION OF AI (17TH CENTURY TO 1930)

The period spanning from the mid-1600s to the 1930s is often referred to as the "incubation phase" of Artificial Intelligence. It was during this era that foundational ideas of modern computing began to take shape, notably through Charles Babbage's proposal of the Analytical Engine. Ada Lovelace, while evaluating Babbage's design, pointed out its limitations by stating that "the Analytical Engine has no pretensions whatever to originate anything. It can do whatever we know how to order it to perform. It can follow analysis; but it has no power of anticipating any analytical relations or truths."[15] Interestingly, this observation later came to be associated with the broader conceptual understanding of AI.

Several pivotal events occurred during this phase that are now viewed as significant milestones in AI's evolution. One such instance is Jonathan Swift's satirical work *Gulliver's Travels* (1726), where he described a fictional machine capable of enabling even the most unlearned person to produce scholarly works in fields

[14] Geneviève Fieux-Castagnet and Gerald Santucci, 2019, Artificial Intelligence: A Civilizational Challenge, A Generational Duty, Big Tech Géopolitique, Sep

[15] Ada Lovelace, 1980, Notes on Notes and The Ladies Diary, Yale University: Computer Science, available from http://www.cs.yale.edu/homes/tap/Files/ada-lovelace-notes.html, [18] Gil Press, 2016, A Very Short History of Artificial Intelligence, Forbes Online.

such as philosophy, law, mathematics, politics, and theology.[18] This machine was seen as a parody of Ramon Llull's combinatorial logic and foreshadowed ideas later related to AI.

Further advancements followed, such as Nikola Tesla's demonstration in 1898 of a radio-controlled boat with autonomous capabilities, which is now regarded as a forerunner to modern robotics. In 1921, Karel Čapek introduced the term "robot" in his play *Rossum's Universal Robots*, contributing significantly to the conceptualization of intelligent machines. Inspired by developments in radio

technology, the Houdina Radio Control Company unveiled a driverless car in 1925, operating through remote control mechanisms.

The portrayal of the robot Maria in Fritz Lang's 1927 sci-fi film *Metropolis* also marked a cultural milestone by imagining humanoid machines with complex behavior. Soon after, in 1929, Japanese scientist Makoto Nishimura created *Gakutensoku*, a robot capable of moving its head and hands, altering its facial expressions, and even learning from its surroundings—traits now closely associated with AI.[16]

Although the terminology of Artificial Intelligence had not yet emerged during this timeframe, the core principles and experimental developments laid a strong foundation for its formal conception in the decades to come.

2.2.2 THE BIRTH OF AI (1940 TO 1950)

The second stage in the evolution of Artificial Intelligence—following its incubation and early conceptual development—is often described as the "birth phase" of AI. This era was characterized by a non-linear trajectory of growth: at times progress surged forward, only to slow down or stagnate before advancing again. The catalyst for this dynamic period was World War II, which significantly accelerated technological innovation and fostered interest in integrating human cognitive abilities with machine functionality.

A pivotal figure during this time was Alan Turing, often referred to as the father of Artificial Intelligence. His landmark question, “Can a machine think?” posed in his seminal 1950 paper *Computing Machinery and Intelligence*, sparked global discourse on the possibility of intelligent machines. This marked a turning point in AI research and helped establish its intellectual foundation.

Several key advancements occurred during this phase. In 1939, John Vincent Atanasoff and his assistant Clifford Berry developed the Atanasoff-Berry Computer (ABC), a pioneering machine capable of solving 29 linear equations simultaneously—an essential step in computational problem-solving. In 1943, Warren McCulloch and Walter Pitts introduced a mathematical framework to model biological neurons in their work *A Logical Calculus of the Ideas Immanent*

in Nervous Activity. This model laid the groundwork for neural networks and, eventually, modern deep learning techniques.In 1949, Edmund Berkeley published *Giant Brains: Or Machines That Think*, where he

[16] Rebecca Reynoso, 2021, A Complete History of Artificial Intelligence, G2 Online

highlighted the increasing ability of machines to handle vast amounts of data efficiently. That same year, Donald Hebb's *The Organization of Behavior: A Neuropsychological Theory* further supported the concept of neural networks, proposing mechanisms for learning and adaptation based on brain activity.

During the early 1950s, computer science saw further innovation with Claude Shannon's introduction of a chess-playing algorithm, one of the first AI programs of its kind. Meanwhile, Alan Turing proposed the "Imitation Game"—now famously known as the Turing Test—suggesting that if a machine could engage in conversation indistinguishable from that of a human, it could be considered intelligent. This idea provided the first concrete framework for evaluating machine intelligence.

The momentum continued in 1951 with the creation of SNARC (Stochastic Neural Analog Reinforcement Calculator), the first artificial neural network. These early breakthroughs collectively marked the beginning of AI as a formal scientific discipline.[17]

2.2.3 THE BLOSSOM PERIOD OF AI (1950 TO 1970)

Between the 1950s and 1970s, Artificial Intelligence experienced significant growth, with research expanding rapidly and showing promising progress. During this stage, the overly idealistic expectations of AI's early days began to give way to a more pragmatic understanding of what AI could realistically achieve. It was a remarkable period, as computers—once thought capable of only performing basic calculations—began to exhibit behaviours and capabilities that hinted at genuine intelligence.

Leading institutions such as the Massachusetts Institute of Technology (MIT), Carnegie Mellon University, Stanford University, SRI International, and the University of Edinburgh established dedicated AI research laboratories. Many other universities also began investing in AI exploration. The main focus of these efforts wasn't to build machines that mirrored the human brain but to design algorithms that could replicate the logical problem-solving methods used by humans.

Several major milestones during this time helped shape modern AI. In 1952, Arthur Samuel, working at IBM, created the first self-learning program that could play checkers. A few years later, in 1955, Herbert Simon and Allen Newell developed *Logic Theorist*, the first AI program capable of solving mathematical theorems—it successfully proved 38 out of the first 52 theorems in *Principia Mathematica*.

A turning point came in 1956 when John McCarthy organized a conference at Dartmouth College and formally introduced the term "Artificial Intelligence." This marked the official beginning of AI as a distinct field of

[17] SNARC was developed by the Marvin Minsky and Dean Edmunds, See "1951 Reinforcement Learning" in Clifford A. Pickover, Artificial Intelligence: An Illustrated History, From Medieval Al Robots to Neural Networks, Sterling Publishing Co., Inc., New York, ISBN 978-1-4549-3360-1.

study. In 1958, McCarthy further contributed to the discipline by developing LISP, a high-level programming language that would dominate AI research for decades.[18]

In 1959, the term "machine learning" was introduced by Arthur Samuel, gaining recognition through his research paper, *Some Studies in Machine Learning Using the Game of Checkers*, published in the IBM Journal.[19] That same year, Nils Nilsson proposed using logic-based systems to represent knowledge and reason intelligently—an approach that remains foundational to present-day AI applications such as statistical modelling and image recognition.

In 1964, computer scientist Joseph Weizenbaum created *ELIZA*, a program that mimicked conversations between a patient and a psychotherapist, responding to inputs in natural language. Around the same time, Woody Bledsoe pioneered facial recognition technology, designing a system that could identify individuals based on facial structure— an innovation with lasting impact.

Daniel Bobrow, in his 1964 doctoral thesis, presented *STUDENT*, an early example of a computer system capable of understanding and solving algebraic word problems written in natural language. In 1965, the development of DENDRAL marked another leap forward. Created by Edward Feigenbaum, Bruce Buchanan, Joshua Lederberg, and Carl Djerassi at Stanford, this expert system aided chemists in analyzing unknown organic compounds using mass spectrometry data.

Also in 1965, Soviet mathematician Alexey Ivakhnenko began foundational work on multilayer perceptrons, planting the seeds for what would later be recognized as deep learning—a method of machine learning that allows computers to learn complex patterns through layered neural networks. Although the term "deep learning" only gained popularity in 1986, Ivakhnenko's contributions were groundbreaking.

In 1966, Joseph Weizenbaum unveiled *ELIZA* as the world's first chatbot capable of maintaining conversation in English. In 1968, the character HAL 9000 from the film *2001: A Space Odyssey* captured the imagination of audiences and researchers alike, symbolizing the potential—and the risks—of advanced AI.[20]

A major breakthrough came in 1969 with the creation of *Shakey the Robot* by SRI International. Shakey was not only mobile but also capable of autonomous decisionmaking. Rather than executing fixed sequences, it could plan actions in real time to achieve specific goals—setting a precedent for future developments in robotics.

[18] John McCarthy et al, Dartmouth Proposal (1955) 2.

[19] Clifford A. Pickover, 2019, Artificial Intelligence An Illustrated History, From Medieval Al RobotsTo Neural Networks: as Machine Learning 1959, Sterling Publishing Co., Inc., New York, ISBN 978-1-4549-3360-1.

[20] Nils J. Nilsson, 2009, The Quest For Artificial Intelligence, A History of Ideas and Achievements, Stanford University, October, p.255, ISBN: 978052112293

This era of innovation was aptly summarized by Herbert Simon, who famously stated: *"It is not my aim to surprise or shock you—but the simplest way I can summarize is to say that there are now in the world machines that think, that learn, and that create. Moreover, their ability to do these things is going to increase rapidly until—in a visible future—the range of problems they can handle will be coextensive with the range to which the human mind has been applied."*

2.2.4 THE AI WINTER (FROM 1970 TO 1980)

During this period, Artificial Intelligence research encountered considerable setbacks, largely due to diminished financial support and a decline in government interest. This era came to be referred to as the "first AI winter," characterized by a significant slowdown in advancements when compared to the enthusiastic progress of previous decades.

Nonetheless, some notable strides were made in robotics and automation.

In 1970, researchers at Waseda University in Japan developed *WABOT-1*, recognized as the first humanoid robot. It featured a limb control mechanism, basic vision capabilities, and the ability to engage in simple communication. The following year, in 1971, the U.S. Defense Advanced Research Projects Agency (DARPA) initiated the five-year *Speech Understanding Research* (SUR) program under the leadership of Lawrence Roberts, aiming to enhance machine comprehension of human language.

In 1972, Stanford University introduced *MYCIN*, an early expert system designed to identify bacteria responsible for severe infections and to recommend appropriate antibiotic treatments.[21] However, in 1973, British mathematician and physicist Prof. Sir James Lighthill released a critical report stating that AI had yet to produce the groundbreaking results it had once promised. This report played a pivotal role in decreasing government funding and further deepening the AI winter.[22]

Despite the downturn, a few key developments still emerged. In 1976, computer scientist Raj Reddy provided a comprehensive review of Natural Language Processing in his work titled *Speech Recognition by Machine: A Review*, contributing valuable insights to the field. Then in 1979, researchers at Stanford unveiled the *Stanford Cart*, one of the first self-driving vehicles. It was capable of navigating a room independently while avoiding obstacles, showcasing early progress in autonomous navigation technology.

2.2.5 THE REVIVAL OF AI (FROM 1980-1987)

The emergence of voice recognition systems in the previous decade reignited enthusiasm for advancements in Artificial Intelligence. This resurgence encouraged researchers to explore fresh avenues within the domain, breathing new life into AI research efforts. Highlighting a few key innovations from this period is essential to understand its significance.

[21] Edward Feigenbaum, Expert Systems (1988) 5 Al Mag 28, 30.

[22] Pamela McCorduck, Machines Who Think (2nd edn 2004) 295.

In 1981, the Japanese government introduced the *Fifth Generation Computer Project*, a major initiative aimed at developing intelligent computing systems. A year later, in 1982, a breakthrough occurred with the introduction of the *Hopfield Network*—a novel type of neural network that could independently process and learn information without direct human control.[23]

In 1984, AI experts Roger Schank and Marvin Minsky cautioned that AI had reached a developmental plateau and predicted another impending stagnation period, often referred to as the second "AI Winter." However, contrary to their forecast, progress continued for a few more years.

By 1986, another milestone was reached when Mercedes-Benz, under the leadership of Ernst Dickmanns, unveiled an early version of a *driverless car*. This vehicle was equipped with cameras and sensors and demonstrated the ability to navigate empty roads autonomously at speeds of up to 55 miles per hour—paving the way for future advancements in autonomous driving technology.

2.2.6 SECOND AI WINTER (FROM 1987 TO 1993)

During this period, it became increasingly evident that earlier-developed expert systems were confined to handling narrowly defined tasks. Their limitations led to widespread disillusionment, especially among corporations that had initially invested heavily, hoping for broader and more adaptive capabilities. As a result, enthusiasm declined, and many companies began withdrawing financial support from AI research.

On a global scale, skepticism about AI's viability grew. Even the United States Defense Advanced Research Projects Agency (DARPA) shifted its focus away from AI, opting

instead to fund projects perceived as more attainable and practical. Consequently, this era saw only a few major advancements in the field. Some of the notable ones include:

In 1988, Rollo Carpenter introduced an early interactive chatbot named *Jabberwacky*, which was designed to mimic human conversation in an entertaining and natural manner. That same year, IBM created *Candide*, an autonomous machine translation system. The methodology behind it was explained in a seminal research paper titled *“A Statistical Approach to Language Translation.”*

By 1990, AI researcher Allen Newell called for the development of *unified theories of cognition*. These comprehensive models aimed to incorporate aspects such as language, learning, motivation, imagination, and self-awareness, representing the full spectrum of human intellectual capabilities.

[23] Artificial Intelligence: Watch Historical Evolution of AI, Analysis of the three main paradigm shifts in Artificial Intelligence, European Commission , 2020, Joint Research Centre Report, No. 120469, Luxemberg Publication, EU, p.9, ISBN: 978-92-76-18940-4

2.2.7 THE RESURGENCE OF AI (1993–2016)

Between 1993 and 2016, AI research underwent a remarkable revival, marked by groundbreaking innovations and technological breakthroughs. This era is often referred to as the "AI Boom" due to the explosive progress achieved, primarily fueled by the increasing availability of large datasets—commonly known as "big data"—that enhanced AI systems' training and performance.

One of the earliest milestones of this period came in 1994, when *Chinook*, an AI program designed to play checkers, became the first to win a world championship title against a human opponent. In 1997, another monumental achievement occurred when IBM's supercomputer *Deep Blue* defeated reigning world chess champion Garry Kasparov, showcasing the potential of machine intelligence in complex strategic games.[24]

The year 1999 introduced the concept of the *Internet of Things (IoT)*, a term coined by Kevin Ashton. The IoT refers to a network of internet-connected devices capable of communicating and operating autonomously. These smart devices rely heavily on AI algorithms to analyze data, make decisions, and perform functions without human intervention.

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3. CONCEPT OF ARTIFICIAL INTELLIGENCE

Defining artificial intelligence may appear straightforward at first glance, yet it proves to be quite complex. This is primarily because the term "intelligence" lacks a universally accepted definition, and furthermore, there is little agreement regarding any direct equivalence between human and machine intelligence.[25]

At its core, artificial intelligence can be understood as a form of intelligence that is not biological. However, the notion of "intelligence" itself is subject to varying interpretations among AI scholars and experts. Therefore, before delving into the depths of AI, it is essential to attempt to define what is meant by both "artificial intelligence" and "human intelligence"—terms that have puzzled experts in biology, psychology, and philosophy for many years.

3.1. UNDERSTANDING THE TERM "ARTIFICIAL"

In everyday language, the word "artificial" typically refers to something man-made or synthetic, often implying that it is a replica or substitute of something natural. However, artificial products are not inherently inferior; in fact, they are frequently favored for their convenience or utility. For instance, consider a silk flower crafted to resemble a real bloom—it requires no water or sunlight, making it a practical decoration for homes or offices. While it lacks the fragrance and tactile quality of a natural flower, its visual similarity may be

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[24] IBM Research, Deep Blue Technical Report (1997) 8.

[25] Woodrow Barfield and Ugo Pagallo, 2018, Research handbook on the law of Artificial Intelligence, Edward Elgar Publishing Limited, Northampton Massachusetts, USA, ISBN 978 1 78643 904 8.

striking. Similarly, artificial intelligence could be seen as a digital counterpart to human cognition, potentially surpassing it in speed and precision.

3.2. HUMAN GENERAL INTELLIGENCE

Despite extensive discourse, a definitive explanation of human intelligence remains elusive. Experts from various fields have yet to agree on a singular, all-encompassing definition. Most descriptions compare human cognition to computational processes, encompassing skills such as reasoning, problem-solving, adaptation to unfamiliar

scenarios, and learning from experiences. The distinctions between human and machine intelligence are thus often blurred.

One influential definition is offered by Robert Sternberg, who describes intelligence as the mental ability to acquire knowledge from experience, apply reasoning, retain vital information, and adapt to the practical challenges of life. Standardized test questions like numerical pattern recognition (e.g., 1, 3, 5, 7, 9—what's next?) assess our ability to detect meaningful patterns, reflecting our capacity to learn from experience.[26]

3.2.1. KEY CHARACTERISTICS OF INTELLIGENCE

Since the late 1980s, certain core attributes have been widely accepted as markers of intelligent thought. These include the ability to communicate, possess internal and external knowledge, pursue goals, and demonstrate creativity.

Drawing a parallel between human cognition and artificial systems, it can be argued that both function through data processing—humans rely on memory and experience, while machines use neural networks to mimic similar processes and simulate intelligent behavior.

TYPES OF INTELLIGENCE

According to Niccolò Machiavelli, intelligence can be categorized into three forms: the first type understands matters independently; the second comprehends what others understand; and the third fails to grasp information, either individually or through others. The first type is deemed superior, the second acceptable, and the third ineffective.[63]

Artificial intelligence systems reflect these tiers as well. Basic AI follows instructions or replicates observed behavior (first level), more advanced AI interprets knowledge acquired by others (second level), and the most sophisticated systems use internal algorithms to arrive at conclusions, independent of external influence or human logic (third level).

[26] Ben Coppin, 2004, Artificial Intelligence Illuminated, Jones and Bartlett Publishers, Mississauga, Canada, P.4, ISBN 0-7637-3230-3

DEFINING ARTIFICIAL INTELLIGENCE

AI is a subfield of computer science dedicated to enabling machines to perform tasks that require intelligence. While it would be convenient to offer a single, universally accepted definition of AI, the field's expansive nature has led to varying interpretations and ongoing debates among researchers.[27]

Functionally, artificial intelligence can be described as technology capable of analyzing its surroundings and making autonomous decisions to achieve predetermined goals. In theory, it aims to replicate aspects of human cognitive ability. AI exists both in virtual forms (such as voice assistants, facial recognition, translation tools, or spam filters) and in physical applications (like self-driving vehicles, robotics, smart appliances, and military equipment). In everyday life, AI plays a vital role in tasks ranging from language translation to content recommendation.

Some experts view AI as software designed to analyze data and make predictions based on historical input. Others consider it an advanced search mechanism capable of identifying optimal outcomes from numerous possibilities.

Although there is no single, authoritative definition in scientific or legal contexts, numerous attempts have been made to frame what AI entails:

- Oxford Online Dictionary defines AI as "the theory and development of computer systems able to perform tasks normally requiring human intelligence, such as visual perception, speech recognition, decision-making, and translation between languages."
- Cambridge Online Dictionary describes AI as "the use of computer programs that have some of the qualities of the human mind, such as the ability to understand language, recognize pictures, and learn from experience."

While useful, these definitions do not entirely capture the complexity of AI.[28]

HISTORICAL DEFINITIONS BY PIONEERS

John McCarthy, often regarded as the father of AI, coined the term in 1955 during a summer workshop at Dartmouth College. He characterized AI as the development of machines capable of behavior that would be considered intelligent if demonstrated by humans. However, this definition is sometimes viewed as lacking scientific rigor.

Marvin Minsky, another prominent figure in the field, described AI in 1967 as "the science of making machines perform tasks that would require intelligence if carried out by people." Although both definitions are foundational, they do not fully encapsulate the evolving nature of the discipline. Other scholars have proposed more detailed explanations:

- Bellman saw AI as the automation of activities associated with human thinking, such as problem-solving and learning.

[27] Bringsjord, et.al., 2020, Artificial Intelligence: The Stanford Encyclopedia of Philosophy, Summer Edition, Edward N. Zalta (ed.), Metaphysics Research Lab, Stanford University

[28] Cambridge dictionary, 2020, Cambridge University Press, [n.d.], Available from,https:// dictionary.cambridge.org/us/dictionary/english/upcycling

- Charniak and McDermott referred to AI as the study of cognitive functions via computational models.
- Haugeland called it "the exciting new effort to make computers think—machines with minds, in the full and literal sense."
- Kurzweil emphasized its artistic and functional dimensions, calling it "the art of creating machines that perform functions that require intelligence when performed by people."
- Winston defined AI as "the study of the computations that make it possible to perceive, reason, and act."
- Rich and Knight summarized it as "the study of how to make computers do things at which, at the moment, people are better."

Based on the above, it is evident that no single definition fully encompasses AI's vast and dynamic scope. To gain deeper clarity, we may refer to Russell and Norvig's framework (2009), which outlines four primary approaches to defining AI, further discussed in subsequent sections. [29]

3.4 RUSSELL AND NORVIG'S APPROACH TO DEFINE ARTIFICIAL INTELLIGENCE

Russell and Norvig have categorized Artificial Intelligence into four distinct conceptual frameworks to provide a comprehensive understanding of its scope and functionality. A concise examination of each of these perspectives can assist in gaining a more nuanced insight into what constitutes AI. These four frameworks are:

- Acting Humanly, also referred to as the *Turing Test Approach*
- Thinking Humanly, known as the *Cognitive Modeling Approach*

- Thinking Rationally, identified as the *Logical Reasoning or Legalistic Approach* • Acting Rationally, commonly termed the *Rational Agent Approach*

3.4.1. ACTING HUMANLY: THE TURING TEST APPROACH

This initial approach incorporates the idea of the *Turing Test*—a conceptual experiment proposed to sidestep the philosophical complexities of the question, *"Can machines think?"* According to this test, if a human evaluator is unable to distinguish between a machine's responses and those of an actual person during a textual conversation, the machine is said to have successfully passed the test. For this to occur, the machine must demonstrate capabilities such as natural language processing, knowledge representation, robotic movement, logical inference, and speech recognition. These elements collectively enable a system to imitate human-like behavior convincingly.

3.4.2. THINKING HUMANLY: THE COGNITIVE MODEL APPROACH

This approach posits that, for a machine or software to be recognized as thinking in a human-like manner, it is essential first to understand how human beings engage in cognitive processes. Various methods such as introspective analysis, neurological imaging, and psychological experimentation are employed to examine the

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[29] Stuart J. Russell and Peter Norvig, 2021, Artificial Intelligence: A Modern Approach, Pearson Series in Ai, 4th ed., Pearson Education, Inc., p.3, ISBN-13: 978-0-13-461099-3.

human mind. One of the earliest computer programs to showcase this approach was the General Problem Solver (GPS), which attempted to replicate human problem-solving methods by modeling them computationally.

3.4.3. THINKING RATIONALLY: THE LOGICAL OR LEGAL REASONING APPROACH

Rooted in classical philosophy, this perspective is heavily influenced by Aristotle's notion of "correct reasoning," which is anchored in formal logic and irrefutable rules of deduction. A typical example might be: *"x is a lion; all lions are male; therefore, x is male."* This form of structured reasoning is commonly referred to as logic.

However, two significant challenges are associated with this method:

First, informal or incomplete information often cannot be easily converted into the precise format that logical systems require.

Second, there exists a noticeable gap between solving theoretical problems and applying those solutions effectively in practical settings.

Despite these limitations, this logic-driven model holds considerable relevance today, especially in the development of self-learning AI systems, which also rely on foundational principles of formal reasoning.

3.4.4 ACTING RATIONALLY: THE RATIONAL AGENT APPROACH

This perspective conceptualizes artificial intelligence as a "rational agent," aiming to explore and construct agents that consistently make the most appropriate decisions. In this context, an "agent" is defined as an entity that performs actions (the term stems from the Latin word *agere*, which means "to do"), and a "rational agent" is one that acts to achieve the most beneficial outcome, or in cases involving uncertainty, selects the most probable beneficial result.

From this viewpoint, intelligence is attributed to the cognitive ability to derive logical conclusions that lead to suitable actions—sometimes a prerequisite for rationality. One method of acting rationally involves deducing the most effective course of action and executing it. However, rational action is not always dependent on inference. For example, in a football match, an AI-based goalkeeper not only calculates the speed and trajectory of an approaching ball and determines the necessary steps to stop it but also physically performs the required maneuver to block the ball.

Reflecting on this understanding, Stuart Russell and Peter Norvig in 2016 offered a comprehensive definition of artificial intelligence: "Artificial intelligence is the discipline focused on the study and creation of intelligent agents, where an intelligent agent is a system that perceives its surroundings and takes actions that optimize its likelihood of achieving success."

3.5 CONCLUDING DEFINITION OF ARTIFICIAL INTELLIGENCE

The existing interpretation of AI is the cumulative product of over six decades of academic inquiry and technological development since the term was first coined. Nevertheless, the concept remains somewhat ambiguous. Drawing from the aforementioned perspectives, artificial intelligence can be generally described as the pursuit of constructing systems that exhibit intelligent behavior, enabling machines to operate effectively in diverse environments through perceptual capabilities. In broader terms, artificial intelligence encompasses systems engineered by humans that, when given a complex objective, can function within physical or digital environments by perceiving their surroundings, analyzing both structured and unstructured data, deriving conclusions from such data, and choosing optimal actions (as per predefined criteria) to achieve the designated aim. Furthermore, these AI systems may be designed with learning capabilities, enabling them to refine their actions over time by evaluating the effects of previous behaviors on their environment.

In view of this analysis, the researcher proposes that artificial intelligence is an integrated system that mimics cognitive human abilities within machines, allowing them to perform tasks at an efficiency and capacity far exceeding human potential.

4 TECHNICAL ELEMENTS OF ARTIFICIAL INTELLIGENCE

Artificial intelligence functions by combining large-scale data inputs, rapid and repetitive data processing, and highly advanced algorithms. For an AI system to operate effectively, several fundamental components must be integrated, including:

- Algorithms
- Hardware
- Data and Information
- Expert Systems
- Neural Networks
- Fuzzy Logic

The algorithm is the foundational building block of AI systems. The term itself traces back to the Persian mathematician Al-Khwarizmi from the 9th century, whose works laid the groundwork for manual problem-solving using mathematical principles. As per Kaufman, an algorithm constitutes "a series of mathematical directives, a structured sequence of operations intended to deliver a desired result within a specified timeframe."[30] Similarly, Ed Finn characterizes it as "a systematic set of instructions for handling data or navigating a problem," while Cormen describes it as

"a clearly defined computational sequence that accepts input values and generates corresponding output values."

[30] Jyh-An Lee, et.al., 2021, Artificial Intelligence and Intellectual Property, Oxford University Press, Oxford, United Kingdom, p. 11, ISBN 978– 0– 19– 887094– 4.

However, regardless of its computational potential, an algorithm requires a physical medium to function—namely, hardware. Therefore, hardware forms the next essential pillar of artificial intelligence. The operational speed of the Central Processing Unit (CPU) is of particular significance. Drawing an analogy, Hans Moravec once noted that "just as an aircraft relies on horsepower, AI requires high computing speed." Below a certain performance threshold, operations become infeasible; conversely, as processing power increases, previously impossible tasks become achievable. This analogy emphasizes that a machine's speed and memory capacity are vital for executing intricate computations. Moreover, AI systems necessitate access to vast datasets to deliver accurate results. The advent of quantum computing—spearheaded by companies such as Microsoft—promises an exponential leap in analytical capabilities. For instance, in 1997, IBM's Deep Blue processed 200 million possible chess moves per second to defeat Garry Kasparov. In contrast, quantum machines have the potential to evaluate a trillion moves per second.

Another critical element in the effective functioning of AI is data and information. AI relies on datasets to apply algorithms and arrive at decisions. Russell and Norvig highlighted that, initially, AI research predominantly centered on algorithm development. However, contemporary studies have shifted focus towards data, recognizing that meaningful outcomes now largely depend on the size and quality of the data rather than the complexity of the algorithms themselves. As per De Mauro,

"Big data represents informational assets of such immense scale, speed, and variety that their transformation into value requires specialized technological and analytical tools."[31] Such datasets are typically far too voluminous to be processed by conventional computing methods within reasonable time constraints.

Expert systems constitute another integral component of AI. These are specialized computer programs capable of executing tasks that would normally require the expertise of a trained human professional. By leveraging one or more AI methodologies, expert systems demonstrate an expert-level understanding of specific

subjects and utilize that knowledge to deliver appropriate responses. They employ sequential (serial) processing techniques to generate targeted outcomes.

In continuation, neural networks represent a further cornerstone of AI, responsible for enabling intelligent data processing. The first neural network computer, named SNARC, was constructed in 1950 by Marvin Minsky and Dean Edmonds. Advocates of the bottom-up approach to AI endeavor to create digital models that mimic the human brain, composed of billions of interlinked cells, or neurons. Neural networks are particularly effective in tasks involving pattern recognition.

Lastly, fuzzy logic adds a nuanced mathematical layer to artificial intelligence, enabling it to express concepts using natural language rather than precise numerical variables. It facilitates responses based on inputs received from sensors, rather than explicit human commands. This logic permits multiple potential outputs, offering a

[31] Favaretto, M., et.al., 2020, What is your definition of Big Data? Researchers' understanding of the phenomenon of the decade. PloS one, 15(2), e0228987, 25 February

greater degree of flexibility and adaptability. As such, fuzzy logic is particularly valuable when navigating uncertainty and making probabilistic decisions.

5 CONCEPT INVOLVED IN ARTIFICIAL INTELLIGENCE

In the present research, it has been observed that the development and application of artificial intelligence encompass a variety of technical expressions and terms. Understanding the literal and contextual meanings of these terms is essential to fully grasp the nuances of this study. The following section aims to clarify some of these core concepts:

5.1 ROBOTS

While the terms *robotics* and *humanoid* are often used synonymously due to their shared ability to replicate human-like actions with a certain level of independence, they should not be considered identical. A *robot* generally refers to a tangible machine equipped with the capacity to make autonomous decisions. In contrast, *artificial intelligence* pertains to the intelligent software that governs and directs such robotic systems. The functionality of a robot is supported by AI-driven algorithms comprising programming logic, actuators, and various sensors that work in unison to perform tasks.[32]

Importantly, artificial intelligence does not always require a physical manifestation like that of a robot to demonstrate its capabilities. Numerous AI applications operate entirely in digital environments—such as automated email categorization, spam detection in Gmail, Google Translate, the Google search engine, and the personalized recommendation system on Amazon—all of which exemplify AI working independently of robotic frameworks.[33]

5.2 MACHINE LEARNING

Machine Learning (ML), as defined by Arthur Samuel, refers to a system's capability to learn and improve from experience without being explicitly coded for specific tasks. This field involves the application of algorithms to analyze data, derive insights, and subsequently make predictions or informed decisions.

Machine learning currently represents one of the most widely utilized branches of AI. It empowers software systems to recognize patterns in existing datasets and apply this knowledge when presented with new data. Examples of ML-powered applications include virtual assistants like Apple's Siri, chatbots, autonomous driving systems, and personalized content suggestions on platforms like Netflix.

5.3 DEEP LEARNING

Deep Learning is a specialized domain within machine learning, inspired by the complex architecture of the human brain—particularly the interconnectivity between neurons. This technique is characterized by its use

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[32] Alfred V. Aho and Jeffrey D. Ullman, 1987, The Elements of Artificial Intelligence, An Introduction Using LISP, Principles of Computer Science Series, Computer Science Press Inc.

[33] Alex Owen-Hill, 2017, What's the Difference Between Robotics and Artificial Intelligence?', Robotiq, 19 July

of multiple processing layers within artificial neural networks, which is the basis for the term "deep" in deep learning.

In deep learning models, artificial neurons act as foundational units arranged in layered structures, where each layer interacts with others to process data and identify abstract representations. Every layer is responsible for learning different features or attributes from the input. Through this multilayered architecture, machines can learn intricate patterns and hierarchical representations, enabling them to make refined decisions. The process involves assigning appropriate weights to inputs to optimize the final output through a network of hidden layers.

Maryland, p.461, ISBN 0-88175-133-8

CHAPTER 3 PATENT LAW ISSUES IMPACTED BY AI

While considerable scholarly attention and regulatory clarification have been given to the influence of AI on copyright legislation, patent law has not received equivalent scrutiny. A notable example in the realm of copyright is the aftermath of a legal case concerning a monkey that captured a selfie. In response, the United States Copyright

Office issued updated guidance in 2016, refining its interpretation of "authorship" to exclude any works created by automated systems or mechanical processes operating randomly or without human input. It reiterated that copyright protection is reserved for "the products of intellectual effort" stemming from the "creative faculties of the human mind." However, unlike copyright law, no comparable clarification has been provided for patent law, and academic dialogue surrounding AI's implications for U.S. patent regulations remains limited. Considering the rapid pace at which AI is transforming both technology and society, a deeper examination of its potential impacts on patent frameworks is crucial. Doing so would ensure the U.S. patent system continues to fulfill its primary functions while mitigating possible adverse social, economic, and ethical consequences.

1. Patent Eligibility of Subject Matter in the Context of AI

Before delving into more complex and novel areas of disruption—such as the patentability of AI-generated inventions—this section focuses on the ongoing and contentious debate over the eligibility of AI-based software under current patent law. Although the number of AI-related patents granted in the U.S. continues to rise, changes introduced to the criteria for patentable subject matter in 2014 have made the path to patenting such technologies significantly more rigorous. This tightening of legal standards poses considerable obstacles for AI patent applicants. Given the profound societal influence of artificial intelligence—surpassing that of traditional, non-intelligent software—it becomes necessary to critically examine how these elevated standards affect innovation, ethical concerns, and economic development. As Justice Richard Linn of the United States Court of Appeals for the Federal Circuit warned, the consequences of misjudging these issues are especially critical for some of today's most transformative innovations, including those in AI and computing technologies.

2. LEGAL STRUCTURE GOVERNING THE PATENTABILITY OF ARTIFICIAL INTELLIGENCE-BASED INVENTIONS

Under Section 101 of Title 35 of the United States Code (hereinafter referred to as 35 U.S.C. § 101), the scope of patentable subject matter is confined to any "new and useful process, machine, manufacture, or composition of matter, or any new and useful improvement thereof." This provision excludes from patent eligibility those claims that pertain to abstract concepts (such as mathematical formulas), naturally occurring phenomena, or fundamental scientific principles. As stated by the United States Supreme Court, these exclusions are justified because such ideas constitute the essential building blocks of scientific and technological progress, and granting exclusive rights over them could obstruct innovation.[34]

In the notable case of *Alice Corporation Pty. Ltd. v. CLS Bank International*,47 the Supreme Court tightened the criteria for acquiring patents on software and "computerimplemented inventions." The *Alice* decision has since been adopted by the Federal Circuit and various lower federal courts to generally reject claims that describe activities capable of being executed by "an ordinary mental process," "within the human mind," or "by an individual using pen and paper," unless those claims demonstrate a distinct technological advancement beyond tasks typically done manually. Such claims must exhibit an "inventive concept" to be considered patenteligible.

This legal framework has introduced a level of complexity for AI-related patents, especially considering that artificial intelligence is designed to replicate cognitive functions typically performed by humans. For instance, in the case of *Purepredictive, Inc. v. H2O.AI, Inc.*, the United States District Court for the Northern District of California invalidated U.S. Patent No. 8,880,446, which was associated with AIpowered predictive analytics. The court determined that the patent was focused on a mental process involving the use of mathematical algorithms for predictive analysis, rendering it an abstract concept. Moreover, the court noted that the patent claims did not reflect a concrete improvement in existing computer-related technology and were, therefore, classified as ineligible for patent protection.

A similar conclusion was drawn in *Blue Spike, LLC v. Google Inc.*, where the court employed the *Alice* test and found that the claims revolved around the general application of an abstract idea using a standard computer system.[35] The claims were dismissed as an attempt to mimic the natural human capacity to detect and identify signals, something historically achievable without computer assistance.48 Because the claims failed to offer

[34] Robert P. Merges, Peter S. Menell and Mark A. Lemley, Intellectual Property in the New Technological Age (Vicki Been et al. eds, 6th ed., 2012) (citing Mayo Collaborative Servs. v. Prometheus Lab., Inc., 566 U.S. 66 (2012)).

[35] Blue Spike, LLC v. Google Inc., No. 14-CV01650-YGR, 2015 U.S. Dist. LEXIS 119382, at *13-16 (N.D. Cal. 8 September 2015), aff'd in Spike v. Google Inc., No. 2016-1054, 2016 U.S. App. LEXIS 20371 (Fed. Cir. 2016).

any "inventive concept" and encompassed broad, generalized processes long undertaken by humans, the court held them invalid.[36]

These judicial interpretations and rulings have made it increasingly difficult for innovators to secure patents for AI-related inventions during the application process and equally challenging for patent holders to maintain the enforceability of such patents during litigation.

3. ANALYTICAL DISCUSSION ON THE CURRENT LEGAL STANDARD

A thorough discussion is necessary to evaluate whether the existing patent eligibility framework effectively supports the primary objectives of the United States patent system. One critical area of analysis is whether the current legal approach facilitates or hinders the advancement of cutting-edge technologies like artificial intelligence. There is a long-standing belief that patent protection acts as a catalyst for innovation, promoting investment and inventive activity. Many suggest that extending patent rights to software encourages research and development within the software sector, thereby furthering technological progress.

Applying this rationale to artificial intelligence is arguably more persuasive, given AI's transformative capabilities surpassing those of traditional software. However, contrasting views argue that software patents may obstruct rather than promote innovation. Some critics advocate for a complete prohibition on software patentability, while others propose restricting the duration of such patents to lessen their potential negative impacts. Moreover, as previously discussed, the judiciary has frequently invalidated patent claims that imitate or mirror human cognitive functions due to the absence of an "inventive concept."[37]

To properly assess the role of AI-related patents in advancing innovation, these diverse viewpoints must be carefully weighed. It is also necessary to consider whether alternative mechanisms, such as trade secret laws or copyright protections, might serve as more effective means of safeguarding AI innovations. Similar analyses are essential in regard to other foundational goals of patent law, including encouraging the disclosure and public dissemination of useful knowledge and promoting the creation of new inventions.

Furthermore, any assessment should take into account AI-specific characteristics rather than relying solely on general considerations applicable to software. The potential impact of AI on society, the economy, and ethical standards could differ significantly from those caused by earlier technological developments. For instance, there are growing concerns that AI may significantly disrupt labor markets, potentially leading to large-scale unemployment106 and more severe economic consequences than those associated with prior innovations. In contrast, some argue that AI's macroeconomic effects will be comparable to historical technological changes. Still, others caution that even if the economic shifts are similar, the effects on inequality and declining labor participation may already be worsening. There is also an argument that

[36] Daniel F. Spulber, "How Patents Provide the Foundation of the Market for Inventions", Northwestern Law & Econ. Research Paper No. 14-14 (26June 2014),

[37] Alice Corp v CLS Bank 573 US 208, 217-218 (2014).

increased support for AI could lead to revolutionary breakthroughs, raising productivity and enhancing global well-being, potentially offsetting any negative implications for employment and income distribution. Additionally, a pertinent policy question arises—would lowering the eligibility threshold for AI inventions disproportionately benefit dominant players in the AI patent landscape? Since AI systems may independently generate novel inventions—a capability beyond traditional software—the competitive advantage of early patent holders in the AI sector might be more substantial than what we see with general software.

Some experts contend that unless reforms to patent law are made soon, these firstmovers may amass disproportionate control and influence, intensifying existing socioeconomic disparities, such as wage inequality. Thus, determining the appropriate legal reforms to enhance the ethical and societal gains of AI is crucial. If changes are warranted, one potential reform could involve lowering the threshold of patentability for AI innovations in fields with high social value—such as medicine, environmental sustainability, criminal justice, and education. Ultimately, stakeholders must engage in comprehensive evaluations to ensure that U.S. patent legislation continues to achieve a fair balance between promoting innovation and addressing competing ethical, economic, and societal concerns.

Patent Eligibility and Inventorship Challenges Concerning AI-Originated Inventions

The issue of whether inventions generated by artificial intelligence qualify for patent protection, as explored in this section, differs distinctly from the discussion surrounding the patent eligibility of innovations aimed at AI systems themselves, which was addressed previously. The central inquiry here is whether creations originating from AI—ideas that would typically be recognized as "inventive" had they been conceptualized by a human—should be acknowledged and safeguarded under existing patent regulations. Additionally, if such inventions are deemed patentable, a critical follow-up question arises: who rightfully qualifies as the inventor? The urgency of resolving these questions is amplified by emerging cases in which patents have already been granted for inventions attributed to AI systems, such as the outputs of the Invention Machine and the Creativity Machine.

1. Legal Implications Related to AI-Based Patentability and Inventorship The foundational philosophy of the United States patent framework is grounded in utilitarian and economic justifications, aiming to encourage the production of novel and beneficial creations. Thomas Jefferson, who served as the country's third President from 1801 to 1809 and played a formative role in the early development of the patent system through the Patent Acts of 1790 and 1793, firmly supported this utility-based approach.[38] He advocated that inventors should enjoy exclusive rights

over their inventions for a limited period, thereby motivating individuals to pursue ideas with potential societal utility. Hence, the US patent regime fundamentally revolves around incentivizing human ingenuity.

[38] Graham v. John Deere, supra note 8, 383 U.S. at 7 ("Thomas Jefferson, who as Secretary of State was a member of the group, was its moving spirit and might well be called the 'first administrator of our patent system

Although the US Patent Act does not specify a fixed threshold for human involvement in the invention process, the language embedded in the law clearly centers on human participation. The concepts of inventorship and patentability under US law are directly linked to human contributors. Patent rights are initially assigned to the inventor, a notion that aligns with prevailing societal values about honoring human intellectual effort and stimulating human-centered innovation. According to US jurisprudence, "conception"—defined as the mental act of forming a definite and enduring idea of a working invention—constitutes a core requirement, and this act must be carried out by a natural person.52 Judicial interpretations by the Federal Circuit have consistently stated that only individual human beings, not entities such as corporations or governments, can fulfill this cognitive function. Similarly, throughout the Patent Act, terminology underscores human agency: Section 101, which outlines what subject matter is patentable, refers to "whoever" invents, while Section 102 on novelty prohibits the patenting of anything not invented by a "person." Additionally, the patent application process mandates an inventor's declaration or oath, reinforcing the necessity of individual authorship.

Restricting patent eligibility to inventions conceived by humans is consistent with the policy of the United States Copyright Office, which likewise does not extend protection to machine-generated works. Yet, this human-centric language in patent legislation may merely reflect the historical context in which the laws were drafted—a time when the notion of machine-originated inventions was not yet technologically relevant. The emergence of AI as a creative force is a relatively recent development, and as such, prior legal frameworks did not anticipate the need to redefine inventorship beyond human boundaries.

To date, neither the US legislature nor the judiciary has directly confronted the legal status of AI-generated inventions. The unresolved question remains whether such innovations should fall within the protective scope of patent law, and if so, who is entitled to claim inventorship in these circumstances.

4. Discussion Points Regarding the Patentability of AI-Generated Inventions

The issue of whether AI-generated inventions qualify for patent protection must be evaluated with reference to whether such protection advances the core objectives of patent law. Some experts argue that acknowledging patent rights for creations made by AI would significantly boost innovation, possibly leading to breakthroughs unattainable through human efforts alone.[39] On the contrary, another school of thought believes that patents do not inherently stimulate innovation, regardless of whether an invention is crafted by a human or an AI system. According to this viewpoint, the proliferation of patents driven by AI could raise barriers to entry, inflate societal costs, create monopolistic structures, and ultimately hinder innovation. In line with this, China's New Generation Artificial Intelligence Development Plan includes language promoting the development of intellectual property rights related to AI, which some interpret as encouraging legal protection for AI-generated outputs, though it stops short of advocating for AI to be recognized as inventors. Others caution that, while AI patent protection might foster innovation, it could simultaneously displace human inventiveness. There are concerns that heavy reliance on autonomous systems for creative tasks might diminish human intellectual engagement, risking the disappearance of high-value R&D roles and entire sectors rooted in human

[39] Burroughs Wellcome Co. v. Barr Labs., Inc., 40 F.3d 1223, 1227-28 (Fed. Cir. 1994).

creativity. Some critics go further, proposing a complete rejection of patent rights for AI-generated works. Instead, they advocate alternative mechanisms such as first-mover advantage, public recognition of AI-generated innovation, or the use of tech-based infringement deterrents to encourage knowledge sharing and technological growth.

Such divergent perspectives demand careful evaluation to understand the overall influence of patent rights on AI-led innovation. It is equally important to determine whether such patents promote the sharing of knowledge and effectively incentivize the appropriate parties—or "beings"—to continue generating useful inventions. Furthermore, the exploration of compromise solutions is essential to reconciling these opposing positions. One proposal is to increase the threshold of patentability—such as elevating non-obviousness standards—for inventions created entirely by AI. This adjustment could establish a fairer competition between human inventors and intelligent machines. Alternatively, differentiated patent terms could be assigned based on the extent of human contribution involved in the inventive process. These

suggestions would require safeguards to ensure that applicants truthfully disclose AI involvement and do not exploit the system.

The ethical, societal, and economic aspects of AI innovation must also be weighed. One avenue might involve tightening the utility requirement under 35 U.S.C. § 101, such that only AI inventions with significant usefulness qualify for patent rights. Historically, the moral utility doctrine was used to deny patents for socially objectionable items like gambling devices; similar reasoning could be adapted to AI. Another option could be limiting patent protection to AI-generated inventions that benefit society in domains like public health, environmental sustainability, criminal justice, and education. A related idea is to impose a heightened standard of nonobviousness for AI inventions that do not fall within these categories.

Crucially, any proposed solution must account for human oversight. Granting full autonomy to AI in the inventive process, without any human control, may yield unpredictable and potentially harmful consequences. The debate must, therefore, consider how human accountability can be preserved, promoting transparency, safety, and ethical use in the evolution of AI patent law.

Points of Discussion on Determining Inventorship in AI-Generated Inventions If inventions entirely produced by artificial intelligence become eligible for patent protection, the logical next consideration is identifying who should be designated as the inventor. As outlined earlier in Section III.B.1, the current legal framework stipulates that "conception," or the mental formation of a complete and functioning idea of an invention by the inventor, is a prerequisite for granting inventorship.

Therefore, if this conceptual process occurs entirely within the "mind" of an AI system, there would be no human to legally list as an inventor under existing statutes. This scenario presents two fundamental options: (1) assign inventorship to the AI system itself, or (2) omit the listing of any inventor on the patent document.

Some experts contend that if the AI's contribution qualifies as inventive, then not only should computationally created inventions be considered patentable, but AI itself should also be acknowledged as the inventor. This would align with the constitutional intent behind granting patents. However, this recognition would necessitate granting legal entity or personhood status to AI, a provision not currently accommodated within U.S. legal frameworks. Still, the broader interpretation of a "legal person" — one who holds rights and duties under the law — might extend to include AI, provided the AI's role as an inventor carries legal responsibilities. In theory, if legislators are prepared to take such a step, both inventorship and legal personhood could be extended to AI.

Nonetheless, it is important to critically examine whether such a move would offer tangible improvements to the current patent system. With the exception of Artificial General Intelligence (AGI) or hypothetical superintelligent AI capable of consciousness — neither of which currently exists — most AI entities lack any desire for recognition or motivation to secure patents. These systems, like the Invention Machine or Creativity Machine, can generate innovations regardless of inventorship incentives. This raises a key question: does attributing inventorship to AI deliver any meaningful advantage beyond merely allowing AI-created inventions to be patented? This brings us to the alternative option: omitting the listing of any inventor altogether. Under current legislation, a named inventor is mandatory, but patent laws could potentially be revised to accommodate AI-generated inventions without attributing inventorship. In this model, it becomes essential to maintain proper incentives for individuals who develop and manage the AI systems responsible for inventive output, to ensure that their work continues. While the owner of the AI system will likely appear as the assignee on the resulting patent — thereby having their interests formally represented — the same cannot be said for individual engineers or developers. These contributors, despite playing a vital role, often remain unacknowledged on the patent record.

If this gap in recognition grows and starts hindering innovation, it might become necessary to introduce a new category within the patent system specifically for AI developers, allowing their efforts to be acknowledged. Ultimately, regardless of whether AI is formally recognized as an inventor or no inventor is listed at all, it is imperative to evaluate how each decision could influence the pace of innovation, while also considering its potential ethical and economic implications.

5. **Liability Concerns Related to Patent Infringement by Artificial Intelligence** One of the pressing concerns in patent law arising due to advancements in AI technology is identifying who bears the responsibility when AI systems violate patent laws. As contemporary AI systems have acquired the technical capacity to infringe upon patented inventions, the matter of liability becomes central. Similar to the inventorship dilemma, the pertinent question is whether accountability should rest with the AI itself, the individual utilizing the AI, or the developers responsible for its creation. Another related issue is the methodology for assigning liability in such cases.

1. **Existing Legal Principles Governing Patent Infringement Responsibility**

Patent law safeguards an inventor's right to prevent unauthorized usage of their patented creations in return for public disclosure of their innovations. In accordance with U.S. law, infringement arises when someone, without proper authorization, produces, uses, sells, or imports a patented invention within the country during the patent's validity. Determining whether infringement has occurred involves a two-step analysis: firstly, the interpretation of each term within the claim, and secondly, proving that the accused technology fulfills each claim element—either precisely or under the doctrine of equivalents. U.S. law also acknowledges induced infringement, holding liable any party that knowingly aids another in directly infringing upon a patent. Once infringement is confirmed, the infringing party must compensate the patent owner, typically through monetary damages, and may be subjected to legal injunctions.[40]

Currently, U.S. statutes do not contemplate scenarios where AI independently commits infringement without human oversight. There are no clear directives on how to assign accountability or damages in cases of AI-initiated infringement. However, the European Parliament's resolution dated 16 February 2017, concerning Civil Law Rules on Robotics, offers interpretive assistance. It asserts that AI cannot be held directly liable for any act or omission resulting in third-party harm. Consequently, responsibility must be traced to a human entity—whether a manufacturer, operator, owner, or user—who might have anticipated and prevented the AI's wrongful conduct. Nevertheless, the advancement of AI in terms of autonomy and cognitive functioning raises the crucial issue of whether current legal frameworks are adequate, especially in situations where the infringement cannot be linked to a particular individual.

2. Examination of Patent Infringement Liability Issues

The need to deter patent infringement, regardless of whether it is committed by a human or an AI, is largely undisputed. However, ignoring the attribution of liability in cases involving AI could promote misuse. Therefore, more comprehensive dialogue is required regarding the identity of the liable party and the criteria for assigning such liability. The chosen approach must support the core objectives of patent law and deliver maximal ethical, societal, and economic value.

The European Parliament's resolution presently advocates for holding human actors accountable rather than AI. One proposed liable party is the AI user. Under traditional legal doctrine, a product user is responsible for any damage their usage may cause, and this concept could be extended to cover damages caused by AI. However, this may lead to confusion and reluctance among users, particularly individuals who lack the technical ability to foresee such infringements. In practice, patent holders more frequently pursue developers or vendors of infringing technologies rather than individual users. Even when users are implicated, they often receive indemnity protection from the product manufacturer.[41]

This underscores the alternative proposition of assigning liability to the developer or producer of the AI system. In patent litigation, it is common to hold the creator of a product accountable for infringement. In the context of AI, developers are best equipped to predict and avoid violations and are typically the ones gaining economic benefit from AI commercialization. Furthermore, in product liability law, manufacturers are often

[40] Vishal V. Khatri et al., "Catch Me If You Can: Litigating Artificial Intelligence Patents", Jones Day (De- cember 2017)

[41] Gaia Bernstein, "The Rise of the End User in Patent Litigation", B.C.L. Rev. 55(5) (2014), 1443

held responsible for defects, a concept that could be extended to AI's infringing behavior, treating it as a form of product malfunction.

Nevertheless, in scenarios involving truly autonomous AI, it becomes questionable whether a human agent can reasonably predict or control the AI's actions to prevent infringement. Holding individuals accountable for the unforeseen consequences of autonomous systems may discourage further AI development due to fear of liability, thereby impeding innovation. In such cases, conventional legal doctrines may fall short of effectively allocating liability, particularly when no responsible party can be clearly identified.

One prospective solution is the introduction of compulsory insurance coverage, similar to auto insurance, to cover risks associated with AI's actions. Such a model would need to account for all potential actors in the AI deployment chain. Supplementing the insurance scheme with a compensation fund could ensure coverage even in instances lacking insurance support.

An alternative approach could be to assign liability to AI itself, provided it is granted legal personhood. As discussed earlier, the legal definition of personhood could potentially be extended to encompass AI. Assigning such status may be more justifiable in the context of liability than inventorship. The European Parliament has acknowledged the potential requirement for bestowing legal identity on robots, proposing a classification of "electronic persons" for highly autonomous AI agents, enabling them to bear legal responsibility for damages caused by their actions. Once the responsible entity is identified, it is essential to establish fair principles for calculating liability. The resolution emphasizes that forthcoming laws should not limit the types of damages recoverable merely because the cause was a non-human entity. If a human is held accountable, liability should correspond to their level of input and the AI's degree of autonomy. Thus, the greater the AI's capacity for learning and independent decision-making, the higher the responsibility attributed to its trainer. The resolution also considers a strict liability framework or a risk-based model, subject to thorough analysis. However, some argue against imposing strict liability on developers due to its potential injustice.

Others propose a contractual strategy, wherein parties delineate liability provisions (such as indemnity clauses) within their AI usage agreements to manage risks proactively. Ultimately, these approaches must be evaluated by balancing ethical, economic, and societal impacts, ensuring that legal frameworks remain transparent, fair, and predictable for all stakeholders.

Points for Discussion on Defining a "Person of Ordinary Skill in the Art"

With the increasing integration of artificial intelligence across multiple sectors, it becomes essential to reassess whether the current interpretation of a Person of Ordinary Skill in the Art (POSITA) remains sufficient—especially considering it traditionally refers to a human and not a machine. This raises the question of whether the definition should evolve to include individuals who utilize AI tools in fields where such usage has become standard. If the definition were expanded to include the involvement of AI, the threshold for non-obviousness would be raised significantly. An overly stringent standard could potentially block meritorious innovations from receiving patent protection, thereby stifling advancement. Conversely, a low standard might trigger a surge in subpar patents and increase litigation—especially by patent trolls—against genuine innovators, hampering both commercial activity and economic progress.

Some advocates of redefining POSITA—to mean a human using AI or even AI alone—assert that, as machines capable of generating inventive ideas continue to advance, they raise the benchmark for what qualifies as patentable. This could mean only groundbreaking inventions would receive protection. However, this shift might disadvantage human-originated inventions, limiting their patent eligibility and bringing with it various consequences, as discussed earlier in Section III.B. If AI attains superintelligence, using it as the benchmark POSITA might result in nearly all novel human ideas being considered obvious from the AI's standpoint. Some legal scholars even argue that conventional patent systems may no longer be relevant and suggest that other incentive models—outside of patent law—should play the role of filtering out non-obvious inventions. Any future discussions about redefining POSITA to incorporate AI must thoughtfully assess the trade-offs and implications of such a change.

Additional Considerations

This section delves into other aspects of patent law that might be affected by AI, warranting closer analysis by lawmakers and policy experts. For instance, what should be the legal stance on patent applications completely drafted by AI systems? If systems like Cloem, AllPriorArt, or Specifio evolve to the point where they can independently conceptualize inventions and generate comprehensive applications, should there be specific regulations to govern such activity? While patent regulations typically focus on the originator of the inventive concept rather than the person or system drafting the application, it's worth debating whether a lack of rules on AIgenerated applications could disrupt the existing system.

If AI tools begin submitting a massive volume of patent filings, it could overwhelm the United States Patent and Trademark Office (USPTO), which currently lacks the infrastructure to handle such volume. This problem might be partially mitigated by increasing filing and administrative fees to deter low-quality submissions. Another pertinent issue involves determining whether content created by AI should be accepted as prior art. If it is, this could complicate the USPTO's efforts to identify relevant prior art, making the examination process more challenging. Moreover, the current obligation for applicants to disclose all material prior art known to them may become harder to fulfill when dealing with vast amounts of AI-generated information. Although excluding AI-generated prior art for practical reasons might seem efficient, doing so could contradict the foundational policies behind existing patent disclosure and examination standards. A careful balance is necessary between practical administrative limitations and the legal principles intended to ensure that the best prior art is considered during patent review.

CHAPTER 4 AUTHORSHIP OF WORKS CREATED BY ARTIFICIAL INTELLIGENCE

1. INTRODUCTION

Copyright exists to protect "original works of authorship," granting rights to the individual responsible for the creation. This encompasses works of literature, music, drama, art, and other recognized creative expressions.

While many copyright statutes intentionally leave the definition of "creation" openended, numerous AI-generated outputs often fall outside the boundaries of this protection due to their inability to satisfy key criteria. Legal complications also arise from the interplay of international treaties, domestic laws, and Copyright Office rules concerning the registration of works fully created by autonomous AI systems.

This raises a crucial question: Can a work entirely generated by AI, without human creative input, qualify for copyright protection? If not, should such works be deemed part of the public domain, free from copyright restrictions?

This chapter delves into these concerns, seeking to determine how authorship of AIcreated works should be approached. It scrutinizes the conceptual and legal debates surrounding "authorship," emphasizing the traditional requirement for human creators and exploring whether legal personhood should be attributed to AI. Further, it outlines possible frameworks for assigning authorship to AI-created content to reduce the uncertainties within copyright law.

2. DEFINITION OF AUTHOR

The terms "creator" and "work" are inherently interconnected. Within intellectual property law, the idea of authorship cannot exist without an identifiable creation. Likewise, authorship only becomes relevant when a tangible work is established.

Although defining "author" is complex, it is generally accepted that the author holds the proprietary rights to the work. Most jurisdictions designate the status of author to the individual who originates a copyrightable creation.

In cases where a work is produced within the scope of employment, the employer typically assumes authorship rights. Moreover, when a work meets the criteria of originality and is fixed in a tangible medium, the human creator becomes entitled to copyright protection.

3. REQUIREMENTS FOR AUTHORSHIP

Under copyright law, particularly when dealing with works produced using AI, two essential conditions must be satisfied to claim authorship:

- The author must be a natural person.
- If the author is non-human, a framework for legal personhood must be considered.

4. THE NECESSITY OF HUMAN AUTHORSHIP

There are two legal approaches to dealing with content created with minimal or no human involvement. One option is to exclude such works from copyright eligibility; the other is to assign authorship to the human who developed the AI program.[42]

While no laws expressly prohibit granting copyright to AI-generated works, most countries refrain from doing so, as copyright regimes typically demand a human creator to claim protection.

[42] Hristov Kalin, 2016, Artificial Intelligence and copyright dilemma, IDEA: The IP Law Review, vol. 57, no. 3

4.1. PURPOSE OF HUMAN AUTHORSHIP REQUIREMENT

The fundamental rationale behind requiring human authorship is to foster economic incentives that promote AI industry growth. Ensuring the dissemination and legal safeguarding of AI-generated works depends on human actors.

Since AI machines do not seek compensation, the individuals or institutions investing time, expertise, and funding in their development should receive the legal and economic rewards. AI technology, if not for the creators and sponsors, would not be accessible to the public.

Thus, for assigning accountability and financial benefits related to AI-generated content, the presence of a human intermediary remains essential.

4.2. . INTERNATIONAL LEGAL STANDARDS

Most global copyright treaties implicitly support the notion that only humans can be authors. For instance, while the Berne Convention advocates for the protection of authors, it refrains from defining who qualifies as an "author."

Although most international frameworks do not explicitly mandate that authorship be human, the legislative intent behind them suggests that only natural persons were

ever envisioned as authors—given that machines were not foreseen to possess creative faculties.

To this day, the majority of AI systems still depend on some level of human guidance. Hence, AI is viewed as a creative tool rather than an autonomous originator.[43]

Such AI-assisted outputs, being products of human-machine collaboration, merit copyright protection. Denying this protection could discourage creators from using AI tools. Therefore, it is vital to evaluate the relevant clauses in international conventions and treaties to better understand their applicability to AI-related authorship issues.

CHAPTER 5 OWNERSHIP OF AI-GENERATED WORKS

1. INTRODUCTION

We currently reside in the era of digital innovation, a time when individuals possess unprecedented freedom to express themselves. In previous generations, such an extensive platform for the creation and dissemination of diverse forms of expression was largely absent. This epoch is often referred to as the "digital information era" due to its capability to produce, replicate, modify, and widely share information. However, this same ability also introduces motivations and avenues for intellectual property violations, leading many to label it the "age of infringement."

The rapid pace at which content can be duplicated and distributed via emerging technologies has rendered it increasingly difficult to enforce copyright protections. For instance, innovations like the Kindle and Nook

[43] Arthur R. Miller, 1993, 'Copyright protection for computer programs, databases, and computer

have revolutionized traditional publishing by turning conventional book readers into digital consumers. Although access to content has become easier, this has simultaneously led to a spike in copyright violations, thereby undermining the value of protected works and compelling creators to seek fair compensation.

While Artificial Intelligence (AI) was previously not widely utilized in creative sectors due to its limited capabilities, this trend is now shifting. A prominent example is AIVA (Artificial Intelligence Virtual Artist), a neural network developed by AIVA Technologies based in Luxembourg and London. It has been trained to compose classical music and has successfully received copyright recognition under its own name by the French and Luxembourg Writers' Right Association (SACEM), thereby being legally acknowledged as a composer.[44]

Such instances underscore the growing role of AI in artistic domains. These examples demonstrate AI's capacity to autonomously learn and execute creative tasks without human input. AI can independently prove mathematical theorems and engage in

artistic creation, such as composing poetry, generating musical pieces, and painting original artworks. It is also capable of producing literary outputs like stories, lyrics, visual art, scripts, and translations. In doing so, AI amplifies the utility of existing applications and drives them to their full potential.

Consequently, AI has become a cornerstone of technological progress in numerous developed nations. Its applications extend from assisting legal systems and healthcare sectors to producing artworks and musical compositions. This expansive evolution has deeply impacted society at all levels, including intellectual property rights. No societal segment appears immune to the influence of AI, and this trend is expected to persist in the future. The intersection of AI and IP has thus emerged as one of the most complex and discussed subjects among legal experts and scholars today.

Recognizing this, global organizations such as the World Intellectual Property Organization (WIPO) have acknowledged the necessity to adapt existing legal structures to align with these technological advancements. In 2019, WIPO initiated the "WIPO Conversation on Intellectual Property and Artificial Intelligence," aimed at fostering dialogue among member nations about the implications of AI on IP law. These conversations have explored key areas such as authorship, originality, ownership, infringement, duration of rights, and related concerns. Nevertheless, consensus on these issues remains elusive, as confirmed by WIPO Director General Francis.[45]

[44] Jeremy A. Cubert and Richard G.A. Bone, 2018, Research handbook on the law of AI, EdwardiElgar Publishing Limited, Massachusetts, USA, p.423, ISBN- 978178643904

[45] Tim Collins, 2018, Sophia the robot's creator, Dr David Hanson, claims humans will marry life- likedroids by 2045, Mail Online, 24 May

Inspired by WIPO's lead, various national and international IP bodies—including the European Patent Office, the UKIPO, the USPTO, and India's IP authorities—have begun exploring the challenge of assigning copyright to AI-generated content. An example from India is the registration of an AI-generated painting titled "Suryast" by a machine named Raghav, which was co-authored with human creator Mr. Ankit Sahni. Similarly, Saudi Arabia made headlines by granting citizenship to a humanoid robot named Sophia. These developments reflect the growing global awareness and legal recognition of AI's contributions.

Notably, as AI continues to play an increasingly central role in IP-related domains— particularly in copyright—it becomes crucial to update and enhance legal frameworks, including foundational treaties such as the TRIPS Agreement. For this reason, this chapter is titled "Ownership in AI-generated Work," aiming to examine how Indian copyright law compares with that of other countries, with the goal of offering constructive solutions.

Ownership remains a critical factor for the future of copyrightable works. If a creation is eligible for protection but lacks a legal owner due to statutory gaps, it loses its purpose—becoming a property without an owner, which is legally unsustainable. Hence, defining ownership is indispensable for the recognition and protection of intellectual property.

This chapter further evaluates whether AI-generated content meets the prerequisites of copyrightability under Indian law and seeks to identify the rightful owner of such creations. The chapter draws from legal principles and policies across different jurisdictions, including the European Union's copyright system, the UK's Copyright,
Designs and Patents Act (CDPA), the United States Code (USC), and India's Copyright Act of 1957. It aims to interpret these laws in a way that addresses the emerging issues posed by AI in the copyright arena.

2. OVERVIEW OF OWNERSHIP

2.1 GENERAL CONCEPT OF OWNERSHIP

The notion of ownership has consistently been a subject of intense discussion within legislative frameworks and judicial interpretations. Much of societal discord centers on the matter of ownership, which can pertain to both physical assets and non-physical assets such as intellectual property. Thus, an exploration of intellectual property ownership becomes particularly significant.

Ownership, in its broadest legal interpretation, implies the right of possession, which may be held by an individual, a collective, a corporate entity, or the state. The object of such possession can include land, movable items, or forms of intellectual property like trademarks, patents, and copyrights.

According to Austin, ownership constitutes "a right indefinite in point of user, unrestricted in point of disposition, and unlimited in point of duration over a determinate thing."

In contrast, Salmond defines ownership as "the right to the general or residuary uses of a material object after deducting all special and limited rights of use vested in others through encumbrances."

From a jurisprudential standpoint, ownership can be interpreted as the control an individual exercises over a property or object, which grants them a set of entitlements. Ownership thus serves as a foundational concept from both legal and social perspectives.

2.2 OWNERSHIP UNDER COPYRIGHT

Given that this research revolves around the concept of ownership specifically within the copyright domain, it becomes essential to understand it from that perspective.

The phrase "ownership of copyright" is a cornerstone in copyright jurisprudence, conferred by statute upon the copyright holder. Unlike ownership of tangible goods, the implications of copyright ownership are unique. By default, the author of a work is considered the first copyright owner. The prefix "first" implies that the law recognizes situations where authorship may be transferred or overridden by exceptions—something not commonly found in the ownership of physical goods.

Factors such as the type of work and the context in which it was created, especially in employer-employee relationships, affect who legally owns the copyright. In many cases, works have multiple copyright owners, who share rights equally unless otherwise specified in a contract. Copyright can also be transferred or assigned to another party and can be governed by various legal arrangements such as contracts for publication, research, or funding. These complexities distinguish copyright ownership from the more straightforward notion of tangible property ownership.

2.3 FEATURES OF OWNERSHIP

Ownership typically encompasses three key jurisprudential components:

1. Unlimited
2. Unrestricted
3. Indefinite

2.3.1. UNLIMITED

From the perspective of the copyright holder, ownership is often perceived as perpetual and can be inherited by successors. Nevertheless, this right is not always eternal. It may cease upon the owner's death, bankruptcy, or any other qualifying legal condition.

2.3.2. UNRESTRICTED

Ownership must be free from limitations regarding alienation, meaning the owner should have the liberty to transfer ownership through sale, lease, exchange, or gift. However, the state retains the right to acquire private property for public use, regardless of the owner's consent.

2.3.3. INDEFINITE

An owner has the liberty to use their property in multiple ways, provided such use does not harm others. Legal systems universally recognize this principle, often embodied in the maxim *sic utere tuo ut alienum non laedas*, meaning "use your property in such a way as not to harm your neighbor." Statutory laws may impose such restrictions for the welfare of the community.

2.4. RIGHTS CONFERRED TO OWNER

Legal attribution of ownership entitles the owner to the following categories of rights:

1. Economic Rights ○ Distribution rights ○ Reproduction rights ○ Translation rights ○ Adaptation rights ○ Communication rights
2. Moral Rights ○ Right of paternity ○ Right of integrity ○ Right to retraction

2.4.1. ECONOMIC RIGHTS

Copyright ownership conveys several economic rights that allow the holder to gain monetary benefits from the utilization of the work.

Among these, reproduction stands out as the most fundamental right. A license may be required to authorize such reproduction. Copyright holders can implement business models enabling users to legally copy and use the work.[49]

2.4.1.1. DISTRIBUTION RIGHT

The owner has the authority to distribute the copyrighted material, whether by sale, rental, free use, or gifting. The mode of distribution may vary, and the nature of the right is dependent on specific scenarios.

2.4.1.2. REPRODUCTION RIGHT

Reproduction entails duplicating existing work or modifying it by adding or editing content. This right to copy for commercial gain is exclusive to the copyright owner and unauthorized replication constitutes infringement.

2.4.1.3. RIGHT TO TRANSLATE

Copyright holders possess the authority to translate their works into other languages. Unauthorized translation without permission constitutes an infringement.

2.4.1.4. ADAPTATION RIGHT

Adaptation involves transformation, transcription, or rearrangement of existing copyrighted works. This right typically applies to literary, musical, and dramatic compositions, not to software. Notably, in *Macmillan and Company Ltd. v. K. and J. Cooper*, the court ruled that since the work being copied lacked originality, it did not constitute infringement.

[49] 2 Henrik Uustalo, 2021, A proposition for the protection of works generated by AI, Master Thesis,Tallinn University of Technology, Tallinn, p.45.

2.4.1.5. COMMUNICATION RIGHT

This right pertains to making the work publicly accessible via broadcasting, webcasting, or other means. Unauthorized transmission is a legal violation. In *Indian Performing Right Society Ltd. v. Aditya Pandey*, the Delhi High Court held that a license is mandatory for public performance or communication.

2.4.2. MORAL RIGHTS

Moral rights supersede economic rights as they are tied to an author's personal dignity and the artistic value of the work. In *Amarnath Sehgal v. Union of India*, the court awarded damages of 50 lakhs for the destruction of an artistic work. Indian copyright law enshrines moral rights in Section 57, comprising the rights to paternity, integrity, and retraction.

RIGHT TO RETRACTION

Retraction refers to the process of withdrawing a previously made statement or published work. Under Section 57, an author holds the right to retract their work from circulation if they believe it is harming their reputation. In the case of *Amarnath Sehgal v. Union of India*, the court elaborated on this right, clarifying that retraction allows an author to remove their creation from public access if it is perceived as misrepresenting or tarnishing their image.

RIGHT OF PATERNITY

The paternity right enables an author to assert their identity as the original creator of a work, even after they have transferred its ownership. This right allows the creator to prevent others from falsely claiming authorship of their protected content. A notable case in this regard is *Sholay Media Entertainment Pvt. Ltd. v. Parag M. Sanghavi*, where the court safeguarded the film title "Sholay," compelling the defendant to change his film's title, as it bore a misleading resemblance and threatened the legacy of the original movie.

RIGHT OF INTEGRITY

The integrity right grants the copyright holder the authority to shield their work from distortion or misuse that may harm its reputation. In the matter of *Sajeev Pillai v. Venu Kunnapalli & Others*, the respondent was alleged to have publicized the film prior to its official release. The court ruled in favor of the petitioner, thereby upholding the creator's right to maintain the integrity of their intellectual property.

AI-GENERATED WORKS AND THE OWNERSHIP DILEMMA

Earlier, it was established that neither Indian nor international copyright regimes currently recognize AI as a legitimate author. Nonetheless, two legal concepts— co-authorship and the work-for-hire doctrine—have emerged as practical approaches to defining authorship in the context of AI-generated content. As most jurisdictions deem the author to be the initial copyright holder unless otherwise stipulated, these doctrines offer a viable path for attributing ownership, which is examined further in this section.

DETERMINING OWNERSHIP

When it comes to AI-generated creations, the primary concern post-authorship is identifying the rightful copyright owner. The dilemma arises because copyright law mandates that ownership must vest in a legal person, and AI lacks legal personhood. Consequently, ownership might be attributed either to the AI's developer or to the end user. For instance, if the AI is not sold or licensed and is instead used directly by its creator to produce original content, the developer becomes the nearest human entity and is, therefore, the logical copyright holder.

CHAPTER 6 CONCLUSION AND SUGGESTIONS

CONCLUSION

Artificial Intelligence and Intellectual Property have become indispensable components of contemporary society. AI is now so deeply integrated into the domain of intellectual property that imagining the future development of IP laws without the use of AI seems impractical. The copyright realm, in particular, has seen significant influence due to advancements in AI. In matters of copyright, the identity of the author holds prime importance, as the author is presumed to be the initial proprietor of the copyrighted material. The Copyright Act necessitates that the author be classified as a 'person,' a term that inherently refers to a human being. Although the statute does not expressly use the term "human," various provisions—such as the reference to the author's life span and succession by legal heirs— imply that authorship is tied to a human creator. Nonetheless, there are scenarios where organizations or corporations can be considered authors through the application of legal doctrines like the "work made for hire" principle and the idea of co-authorship. The arrival of artificial intelligence in the copyright sphere has introduced complex challenges regarding the concept of authorship. When AI systems generate creative works, these outputs often satisfy all the criteria set forth under the Indian Copyright Act for protection. However, the requirement that the author be a human individual becomes problematic in this context.

The advanced cognitive and functional capabilities of AI have disrupted the long-held notion that creativity is a uniquely human trait. At present, artificial intelligence allows tasks to be executed with greater ease, speed, and accuracy, leading to widespread adoption of AI-based technologies.

Virtually every sector today has experienced the transformative effects of AI. From composing music and crafting poetry to generating news reports, designing artworks, and drafting architectural plans, AI systems are now capable of outperforming humans in both quality and efficiency. This surge in AI application extends into the intellectual property realm as well, affecting multiple branches. Innovations such as AI-assisted legal professionals, autonomous vehicles, AI-enhanced judicial mechanisms, surveillance infrastructure, and even AI-generated hyperrealistic facial images are just a few examples.

These innovations often align with the requirements for various forms of intellectual property, demonstrating that the influence of AI on this field is both profound and irreversible.

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